Research record: locality, terminal information, and planar exploration laws

Autonomous mathematical investigation, with independent critical checks

26 October 2026

Abstract

The project's three-axiom classification is false under the usual unparameterized-curve interpretation of locality. An exact counterexample chooses an affine SLE_6 metric from the boundary germ of the terminal prime end. This germ survives every operation in the stated axioms. The example can also be covariant under all Euclidean similarities. A strict boundary-swallowing probability identifies an affine metric already from one marked Jordan-domain path law, proving that no fixed affine map normalizes the example. We also prove rigidity for deformations by one global homeomorphism, identify precisely the extra driving-martingale assumption that forces SLE_6, and record domain-continuity, target-consistency, reversal, and near-critical obstructions. Critical percolation universality and classification under suitably strengthened axioms remain open.

Contents

1 Scope, conventions, and verification status **2**

2 An affine metric is identifiable from one chordal law **3**
2.1 A boundary-swallowing race . . . 3
2.2 Boundary rigidity and affine uniqueness . . . 4

3 An exact counterexample using the terminal boundary germ **5**
3.1 The conserved information and the admissible domain class . . . 5
3.2 A concrete example with full similarity covariance . . . 6
3.3 Why these really are continuous-driving Loewner curves . . . 6
3.4 Quasiconformal images have infinite capacity lifetime . . . 7

4 A positive classification within the global-deformation class **8**

5 Boundary stability and other ways to remove terminal dependence **9**
5.1 An explicit failure of continuity under small target perturbations . . . 9
5.2 Reversibility eliminates this mechanism inside the affine-germ ansatz . . . 10

6 Further affine rigidity and target consistency **10**
6.1 Target consistency is a substantive additional condition . . . 11
6.2 A compatibility graph for all target-germ affine decorations . . . 12
6.3 Finite rotations and a fixed anisotropy . . . 12

7 What locality actually says about a semimartingale driving process **13**
7.1 Deletion of a hull and the drift anomaly . . . 13
7.2 A precise conditional classification by driving martingales . . . 14
7.3 A constant-characteristic version and an explicit two-slit test . . . 15

8 Dilation symmetry and criticality **15**
8.1 Dilation covariance and subsequential limits . . . 15
8.2 All-scale RSW and a nonzero near-critical parameter . . . 16

9 Near-critical obstruction: what is rigorous and what is not yet transferred **16**
9.1 An explicit limit-transfer lemma . . . 17

10 Affine identifiability without Jordan boundary regularity **18**
10.1 A quasiconformal disk map fixing the boundary stays close to the identity . . . 18
10.2 The identifiability theorem on any bounded simply connected domain . . . 19
10.3 A necessary and sufficient DMP condition inside the affine ansatz . . . 20

11 Local target consistency passes to marginal curve limits **21**
11.1 Stopping an ordered unparameterized curve on a circle . . . 21
11.2 Transfer using only two marginal limits . . . 22
11.3 Application to two percolation targets . . . 22

12 Affine metrics encoded in initial path germs **23**
12.1 State space and the meaning of a trace germ . . . 23
12.2 Dilation ergodicity . . . 24
12.3 Separating distinct metrics from arbitrarily short initial segments . . . 24
12.4 A common decoder for the continuum of normalized metrics . . . 25

13 A fixed metric follows inside the affine-SLE ansatz with target consistency and reversal **27**

14 Continuous affine labels: deterministic support and source independence **29**
14.1 The precise continuity premise . . . 29
14.2 Full support transfers conservation to deterministic Loewner controls . . . 30
14.3 Rapid driving ramps move the source at vanishing capacity . . . 31
14.4 A fixed metric in the affine ansatz without a reversal hypothesis . . . 32

15 Unsuccessful routes, revised objectives, and next experiments **33**

1 Scope, conventions, and verification status

This record starts from an otherwise empty research repository. The project specification is preserved unchanged. Its primary question is whether strong domain Markov, locality, and physical translation covariance force a domain-indexed family of exploration curves to be one fixed affine pullback of chordal SLE_6. We do not assume this conclusion or conformal covariance of an unknown family.

Curves are oriented, unparameterized continuous paths. After an initial exploration, the remaining domain is the component facing the terminal prime end, with the exposed tip interpreted as an appropriate prime end. Locality compares curves up to first encounter with a removed closed set; their respective capacity clocks need not agree. Deletions must preserve the source and terminal

end neighborhoods and leave a simply connected target component. This is the customary precise setting for chordal locality. For a Jordan target and a planar closed set disjoint from that point, preservation of its terminal neighborhood is automatic.

External facts used. We use existence, conformal covariance, strong domain Markov, locality, and continuous Loewner generation of ordinary SLE_6 as established theorems; see [1]. We use elementary conformal mapping, boundary extension on Jordan domains, harmonic uniqueness, and Morera's theorem. Loewner admissibility of affine images is justified explicitly using [2]. A reversal statement below additionally uses [3]. These external results are distinguished from the arguments proved here.

Proof checks. The target-germ construction was found independently in two investigations. The boundary-race computation and the single-domain metric-identification argument were checked independently, including the sign of the diffusion, the exclusion of simultaneous swallowing, and the boundary-extension step. The global-homeomorphism rigidity argument and the local boundary-twist test were also independently checked. These checks are evidence of correctness, not a replacement for expert human review. Nothing here proves universality for general critical percolation.

2 An affine metric is identifiable from one chordal law

For a bounded Jordan domain D, distinct $p, q \in \partial D$, and an orientation-preserving invertible real-linear map A, put

$$\nu^A_{D,p,q} = (A^{-1})_* \operatorname{Law}\big(\mathrm{SLE}_6(AD; Ap, Aq)\big). \tag{1}$$

Translations in an affine normalizer are immaterial. The tensor $A^{\mathsf{T}}A$ is its Euclidean conformal metric, determined only up to a positive scalar.

2.1 A boundary-swallowing race

Lemma 2.1 (Strict boundary-race probability)**.** *Let η be chordal SLE_6 in $\mathbb{H}$ from 0 to ∞. For $x < 0 < y$, let τ_x, τ_y denote their Loewner swallowing times. Then*

$$\mathbb{P}(\tau_y < \tau_x) = F\left(\frac{-x}{y-x}\right), \qquad F(r) = \frac{\int_0^r [u(1-u)]^{-2/3}\,du}{\int_0^1 [u(1-u)]^{-2/3}\,du}. \tag{2}$$

In particular F is strictly increasing on $(0,1)$. The event in (2) is invariant under orientation-preserving homeomorphisms of marked Jordan domains, when formulated as the order of separation from the terminal prime end by the filled exploration hull.

Proof. Use $\partial_t g_t(z) = 2/(g_t(z) - W_t)$ with $W_t = \sqrt{6}B_t$. Until the first swallowing time put

$$X_t = g_t(x) - W_t < 0, \quad Y_t = g_t(y) - W_t > 0, \quad T_t = Y_t - X_t, \quad R_t = -X_t/T_t \in (0,1).$$

Then

$$dX_t = \frac{2}{X_t}\,dt - \sqrt{6}\,dB_t, \qquad dY_t = \frac{2}{Y_t}\,dt - \sqrt{6}\,dB_t,$$
$$dT_t = \frac{2T_t}{-X_tY_t}\,dt > 0, \qquad dR_t = \frac{2(1-2R_t)}{R_t(1-R_t)T_t^2}\,dt + \frac{\sqrt{6}}{T_t}\,dB_t.$$

With $ds = dt/T_t^2$, the generator of R is

$$\mathcal{L} = 3\frac{d^2}{dr^2} + \frac{2(1-2r)}{r(1-r)}\frac{d}{dr}.$$

The increasing solution of $\mathcal{L}F = 0$, $F(0) = 0$, $F(1) = 1$ has $F'(r) = C[r(1-r)]^{-2/3}$, giving (2) by bounded optional stopping. To justify the terminal values, $|X|/\sqrt{6}$ and $Y/\sqrt{6}$ are Bessel processes of dimension $1 + 4/6 = 5/3$ up to their first zero; thus the first swallowing time is finite almost surely. Alternatively, the accessible endpoints of the displayed diffusion give the same stopping argument. There is no simultaneous swallowing, since $T_t \geq y - x > 0$, whereas both $X_t, Y_t \to 0$ would force $T_t \to 0$. At the first swallowing, $R = 1$ exactly when y is swallowed first. ODE solutions and the continuous driver stay finite up to that first time.

Swallowing means that the marked boundary end has been cut off from the target by the explored filled hull (including the boundary swallowed at first contact). This definition uses only the domain, the oriented path, and its order of traversal. A homeomorphism carries filled target components to filled target components. Capacity parametrization therefore has no role in the comparison event. □

2.2 Boundary rigidity and affine uniqueness

Lemma 2.2 (A single SLE law identifies its conformal boundary map)**.** *Let $h : \overline{D} \to \overline{D'}$ be an orientation-preserving homeomorphism between closed bounded Jordan domains, with $p' = h(p)$, $q' = h(q)$. If $h_*\mathrm{SLE}_6(D; p, q) = \mathrm{SLE}_6(D'; p', q')$, then $h|_{\partial D}$ agrees with the boundary extension of a conformal map $f : D \to D'$.*

Proof. Choose conformal maps $\phi : D \to \mathbb{H}$ and $\phi' : D' \to \mathbb{H}$ sending their marked endpoints to $0, \infty$. Their Jordan boundary extensions give an increasing homeomorphism

$$H = \phi' \circ h \circ \phi^{-1} : \overline{\mathbb{R}} \to \overline{\mathbb{R}}$$

fixing $0, \infty$. For every $x < 0 < y$, equality of the corresponding boundary-race probabilities and strictness in Lemma 2.1 give

$$\frac{-H(x)}{H(y) - H(x)} = \frac{-x}{y - x}, \quad \text{hence} \quad \frac{H(x)}{H(y)} = \frac{x}{y}.$$

Fixing one $x < 0$ and one $y > 0$ shows $H(s) = cs$ for all real s, with $c > 0$. Consequently $f = (\phi')^{-1} \circ (c\phi)$ has exactly the required boundary values. Equality of interior maps is not asserted for an arbitrary h. □

Theorem 2.3 (Single-domain affine identifiability)**.** *For even one bounded Jordan D and one pair of distinct boundary points p, q, the equality $\nu^A_{D,p,q} = \nu^B_{D,p,q}$ holds if and only if BA^{-1} is a Euclidean similarity. Equivalently,*

$$B^\mathsf{T}B = cA^\mathsf{T}A \qquad (c > 0).$$

The statement also allows orientation-reversing normalizers by composing one map with a reflection.

Proof. The equality of laws means $M = BA^{-1}$ carries the SLE law in AD to the SLE law in BD, at the corresponding endpoints. By Lemma 2.2, a conformal map $f : AD \to BD$ has boundary values equal to M. Both real coordinate functions of the real-linear map M are harmonic. The real and imaginary parts of f are also harmonic, continuous on the closure, with the same boundary values. The maximum principle on the bounded domain gives $f = M$ throughout AD. Thus M is complex-linear and invertible, hence a similarity. The converse is ordinary SLE similarity covariance. Reflection covariance reduces the remaining orientation case to this one. □

3 An exact counterexample using the terminal boundary germ

3.1 The conserved information and the admissible domain class

Let $G(D, q)$ be the embedded germ of D at its terminal prime end. Concretely, target neighborhoods may be taken as

$$f^{-1}\big(\mathbb{H} \setminus \overline{B(0, R)}\big), \qquad R \to \infty,$$

for any conformal $f : D \to \mathbb{H}$ sending q to infinity. The embedded germ records the planar domain in a full neighborhood of that end; one may shrink neighborhoods without changing it. A full neighborhood contains a neighborhood in the prime-end compactification, rather than merely an arbitrarily narrow approach sector. Germ equivalence means agreement on such neighborhoods with the same inherited marked end.

At a Jordan boundary point this is the ordinary local boundary germ. At a slit tip one must retain which prime end is intended. A germ is called C^1-regular here when it has a singleton endpoint and, in a full end neighborhood, a single C^1 boundary arc with a nonzero tangent and the domain locally on one side. The tangent line $\ell(G)$ is unoriented. All other germs receive the default label below. Regularity and tangent are properties of the germ itself.

Finite chordal Loewner hulls in a target-normalized half-plane chart are bounded. They therefore miss some target neighborhood. At every finite stopping time prior to reaching the target, the remaining marked domain D_t satisfies

$$G(D_t, q) = G(D, q). \tag{3}$$

Likewise locality deletions away from the terminal end preserve this germ. Thus the following proof already works on domains with regular targets, together with their slit domains, a class closed under all operations needed in the argument. Assigning defaults to nonregular end germs extends it to arbitrary prime ends under the stated end-neighborhood convention. Locality for sets accumulating at a nonsingleton impression must always be formulated in prime-end terms; it is not defined by pretending the end is a single physical point.

Proposition 3.1 (Conserved-label construction)**.** *Assign to each embedded terminal germ an orientation-preserving invertible real-linear map $A(G)$, measurably and in a translation-covariant way: $A(G(D + v, q + v)) = A(G(D, q))$. Define*

$$P_{D,p,q} = (A(G(D, q))^{-1})_* \mathrm{SLE}_6\big(A(G(D, q))D; A(G(D, q))p, A(G(D, q))q\big). \tag{4}$$

Then this family has strong domain Markov, locality, and translation covariance as unparameterized paths.

Proof. Write $A = A(G(D, q))$. By (3) the same deterministic A is prescribed after any finite initial exploration. Apply strong domain Markov to SLE_6 in AD, then pull the tail back by A^{-1}. This gives exactly (4) in the remaining target component, at its exposed tip and target. Stopping times are carried to stopping times because the invertible map preserves the explored filtration; the capacity clock is an adapted increasing time change. For deletion of K away from both ends, the two domains use the same A. SLE_6 locality in AD and $AD \setminus AK$, pulled back by A^{-1}, couples their paths identically until first encounter with K. Finally the displayed covariance of the label and translation covariance of SLE_6 prove physical translation covariance. No conditioning on a hidden random parameter occurs: the label is a deterministic conserved function of the current marked domain. □

3.2 A concrete example with full similarity covariance

Fix $a > 0$, $a \neq 1$, and set

$$A(G) = \begin{cases} \Pi_{\ell(G)} + a\Pi_{\ell(G)^\perp}, & G \text{ is } C^1\text{-regular}, \\ I, & \text{otherwise.} \end{cases} \tag{5}$$

Here Π is orthogonal projection. On smooth-target domains this is a continuous function of the tangent in the topology controlling that tangent; on arbitrary germs it is a deterministic local geometric rule. If desired, the class can first be restricted to piecewise smooth domains and their finite slit descendants, avoiding any measurable-space conventions for the class of all planar domains.

For a similarity $S(x) = cRx + v$, $c > 0$, R orthogonal, one has

$$A(G(SD, Sq)) = RA(G(D, q))R^{-1}.$$

Combining this with SLE_6 similarity and reflection covariance proves $P_{SD,Sp,Sq} = S_* P_{D,p,q}$. Thus adding scale invariance, reflection covariance, or covariance under every rotation does not exclude this construction.

Theorem 3.2 (The proposed fixed-affine classification is false)**.** *The ensemble (4)–(5) satisfies the three proposed axioms, and full Euclidean similarity covariance, but there is no one fixed affine transformation sending all its laws to ordinary chordal SLE_6. This failure already occurs for smooth targets on one disk.*

Proof. Take D to be the unit disk and fix $p = -1$. At $q_h = i$ the tangent is horizontal, while at $q_v = 1$ it is vertical. The respective matrices are

$$A_h = \begin{pmatrix} 1 & 0 \\ 0 & a \end{pmatrix}, \qquad A_v = \begin{pmatrix} a & 0 \\ 0 & 1 \end{pmatrix}.$$

A fixed affine normalizer with linear part B would imply $\nu^B_{D,p,q_h} = \nu^{A_h}_{D,p,q_h}$ and $\nu^B_{D,p,q_v} = \nu^{A_v}_{D,p,q_v}$. Theorem 2.3 makes both BA_h^{-1} and BA_v^{-1} similarities. Hence $A_v A_h^{-1} = \mathrm{diag}(a, a^{-1})$ would be a similarity, forcing $a^2 = 1$, contrary to $a > 0$, $a \neq 1$. □

3.3 Why these really are continuous-driving Loewner curves

Continuity and the absence of transverse crossings alone would not suffice to settle this point. Instead use the deterministic characterization in [2, Theorem 1.1 and Remark 1.4]: a continuous half-plane path with strictly increasing filled hulls is a continuous-driving Loewner trace if and only if it is a locally uniform limit of simple interior paths. The condition is independent of parametrization.

An ordinary SLE_6 trace has this property. Conjugating an invertible affine map by the conformal charts of its source and image domains gives a quasiconformal self-map of $\mathbb{H}$. It extends to a homeomorphism of the closed half-plane with its point at infinity, fixing the two marked ends after normalization. It carries simple interior approximants to simple interior approximants; uniform convergence survives on compact time intervals because the extended map is uniformly continuous on relevant compact sets. Filled hulls and their strict growth are topologically preserved. Yuan's theorem gives a continuous driving function after reparametrizing by capacity. This also treats nonsmooth slit domains in the Markov property without assuming Euclidean continuity of their uniformizing maps on every boundary point. The counterexample therefore applies to the driving-function formulation as well as to curves.

3.4 Quasiconformal images have infinite capacity lifetime

Yuan's characterization gives a continuous Loewner driver after recoding an affine image by its capacity, but this alone does not assert that the capacity lifetime is infinite. A continuous curve tending to infinity need not have infinite half-plane capacity. The following argument proves the required assertion for the quasiconformal changes arising here.

Lemma 3.3. *Let $F : \mathbb{H} \to \mathbb{H}$ be a fixed K-quasiconformal self-homeomorphism whose boundary extension fixes 0 and ∞. If η is ordinary chordal SLE$_6$ from 0 to ∞, then $F\eta$ has unbounded imaginary part almost surely. Consequently its half-plane-capacity lifetime is infinite.*

Proof. Put $C = \overline{B(i, 1/4)} \subset \mathbb{H}$. The boundary map of F is increasing, so $F(R) > 0$ for $R > 0$ and $F(R) \to +\infty$ as $R \to \infty$. The normalized maps

$$F_R(z) = \frac{F(Rz)}{F(R)}$$

are K-quasiconformal half-plane maps fixing $0, 1, \infty$. Reflecting them across the real line gives normalized quasiconformal sphere maps. Compactness of this normalized family, including its inverses, implies

$$\inf_{R>0} \inf_{z\in C} \operatorname{Im} F_R(z) = c_K > 0.$$

Indeed, a sequence violating positivity would have a subsequence converging uniformly on the sphere to a quasiconformal homeomorphism preserving the half-plane, and a subsequence of points in C would then be mapped to its real boundary, an impossibility. Hence

$$\inf_{z\in RC} \operatorname{Im} F(z) \geq c_K F(R) \longrightarrow \infty. \tag{QC1}$$

The event $E = \{\eta \cap C \neq \varnothing\}$ has positive probability. One elementary justification avoids invoking a support theorem. In the normalization $\partial_t g_t = 2/(g_t - W_t)$, the Brownian event

$$\sup_{0\leq t\leq 1} |W_t| < \varepsilon, \qquad 0 < \varepsilon < 1/4,$$

has positive probability. On this event the Loewner hull stays within the vertical strip $|\operatorname{Re} z| \leq \varepsilon$ up to time 1. For example, a point initially to the right of that strip has increasing real part under the Loewner ODE and retains a positive real gap from the driver, so it cannot be swallowed in finite time; the other side is identical. If the entire trace up to time 1 had height at most 1, its filled hull would lie in the half-disk of radius $\sqrt{1+\varepsilon^2}$. By capacity monotonicity its capacity would be at most $1 + \varepsilon^2 < 2$, whereas its actual capacity is 2. Here a half-disk of radius R has hydrodynamic map $z + R^2/z$ and capacity R^2. Thus the continuous trace crosses height 1 in the strip, and that crossing lies in C.

Fix $r > 1$. Brownian scaling, and therefore scalar dilation of ordinary SLE$_6$, is mixing. Indeed $Z_s = e^{-s/2} B_{e^s}$, $s \in \mathbb{R}$, is a stationary Gaussian process with covariance $e^{-|s-u|/2}$; scalar Brownian scaling becomes a log-time shift. Decay of these covariances gives mixing first for Gaussian cylinder functions and then for the whole path sigma-field. Its inverse dilation is ergodic as well. Birkhoff's theorem applied to 1_E gives

$$\eta \cap r^n C \neq \varnothing \qquad \text{for infinitely many positive integers } n, \quad \text{almost surely.}$$

In fact these visits have limiting positive frequency, since $1_E(S_{r^{-n}}\eta) = 1_{\{\eta\cap r^n C\neq\varnothing\}}$. Equation (QC1) then shows that $F\eta$ has unbounded height.

Every finite initial image trace is compact away from infinity and has finite capacity. The standard Loewner height estimate

$$\max\{\operatorname{Im} z : z \in K\}^2 \le 2\,\mathrm{hcap}(K)$$

for a half-plane hull, or equivalently $\max \operatorname{Im} K_u \le 2\sqrt{u}$ when $\mathrm{hcap}(K_u) = 2u$, now forces the capacities of these image prefixes to tend to infinity. Thus the capacity-recoded image has lifetime $[0, \infty)$. □

For an affine image in a bounded marked domain, conjugating the affine map by its source and image conformal charts gives exactly such a fixed quasiconformal F fixing the marked ends. This lemma therefore completes the infinite-time driving-function assertion for the target-germ counterexample. It uses properties of SLE_6, rather than an invalid general assertion about arbitrary curves tending to infinity.

4 A positive classification within the global-deformation class

The preceding example does not arise from one deterministic global change of spatial coordinates. Within that narrower class the proposed affine conclusion does follow, and no differentiability of the change of coordinates is required.

Theorem 4.1 (Full-family homeomorphism symmetries are conformal)**.** *Let $h : \mathbb{C} \to \mathbb{C}$ be an orientation-preserving homeomorphism such that, for every bounded Jordan domain and marked boundary pair,*

$$h_*\mathrm{SLE}_6(D; p, q) = \mathrm{SLE}_6(hD; hp, hq).$$

Then h is an entire conformal automorphism, hence $h(z) = \alpha z + \beta$ for complex $\alpha \neq 0$. For an orientation-reversing homeomorphism, the corresponding conclusion is anticonformality.

Proof. Take any nondegenerate closed Euclidean triangle T and distinct $p, q \in \partial T$. Lemma 2.2 gives a conformal $f_T : T^\circ \to h(T^\circ)$ whose continuous boundary extension equals h on ∂T. Cauchy's theorem, applied on interior triangles and then passed to the boundary by continuity, yields

$$\int_{\partial T} h(z)\,dz = \int_{\partial T} f_T(z)\,dz = 0.$$

Since h is continuous and this holds for every triangle, Morera's theorem makes h entire holomorphic. Its injectivity makes it conformal, and an entire conformal automorphism of the plane is complex affine. Composition with a reflection proves the other orientation case. □

Theorem 4.2 (Stationary global deformations are real affine)**.** *Let $h : \mathbb{C} \to \mathbb{C}$ be a homeomorphism and define, on all bounded Jordan domains,*

$$P^h_{D,p,q} = (h^{-1})_*\mathrm{SLE}_6(hD; hp, hq).$$

If this ensemble is covariant under every physical translation, then h is a real-affine homeomorphism.

Proof. Reflection covariance lets us assume h preserves orientation. For every $v \in \mathbb{C}$, physical translation covariance says that

$$\psi_v = h \circ T_v \circ h^{-1}, \qquad T_v(z) = z + v,$$

preserves the full chordal SLE_6 family. All bounded Jordan domains are available, since inverse images under a plane homeomorphism are bounded Jordan domains. Theorem 4.1 gives $\psi_v(z) = \alpha_v z + \beta_v$. For $v \neq 0$, the conjugate of a nonzero translation has no fixed point. A complex affine map with $\alpha_v \neq 1$ has a fixed point, so $\alpha_v = 1$ for all v. It follows that $h(x+v) = h(x) + \beta_v$, and putting $x = 0$ gives

$$h(x+v) - h(0) = (h(x) - h(0)) + (h(v) - h(0)).$$

The continuous additive map $h - h(0)$ is real-linear. It is invertible because h is a homeomorphism. □

What this does not prove. The original axioms do not give a representation by one global h. The counterexample shows that replacing a domain-dependent geometry by one fixed spatial geometry is a substantive missing step, not an innocent coordinate choice.

5 Boundary stability and other ways to remove terminal dependence

5.1 An explicit failure of continuity under small target perturbations

The tangent rule is continuous when target tangents converge, but not under ordinary uniform approximation of the domains and their boundaries. This defect persists even when every domain is smooth and the approximating domains are uniform quasidisks.

Proposition 5.1 (Local rotation of a terminal germ)**.** *For the ensemble of Theorem 3.2, there are smooth Jordan $D_n \to D$, with uniformly converging boundary parametrizations and fixed distinct marked points p, q, for which $P_{D_n,p,q}$ does not converge to $P_{D,p,q}$ in the uniform unparameterized-curve topology.*

Proof. Let D be the unit disk, $q = i$, $p = -1$. Choose a smooth radial cutoff χ equal to 1 near 0 and 0 outside the unit disk, and define the orientation-preserving smooth diffeomorphism

$$F(re^{i\theta}) = r \exp\bigl(i\theta + i(\pi/2)\chi(r)\bigr).$$

It is a rotation by $\pi/2$ near 0 and the identity outside the disk. Its quasiconformal distortion is finite, since $r\chi'(r)$ is bounded. Put $F_\epsilon(z) = q + \epsilon F((z-q)/\epsilon)$ and $D_\epsilon = F_\epsilon(D)$. Then $F_\epsilon(q) = q$, and $F_\epsilon(p) = p$ for small ϵ, while $\sup_z |F_\epsilon(z) - z| \leq 2\epsilon$. All F_ϵ have one common quasiconformal bound. At q the target tangent has rotated by $\pi/2$, so the matrix used for every D_ϵ is A_v, whereas the matrix for D is A_h.

The domains $A_v D_\epsilon$ are uniform quasidisks converging to $A_v D$, with the marked boundary points converging as well. Kernel convergence and uniform boundary equicontinuity of normalized Riemann maps on uniform quasidisks give uniform convergence of these maps on the closed disk. Coupling their chordal SLE_6 paths with one common disk SLE path therefore gives convergence of the entire unparameterized curves. Thus

$$P_{D_\epsilon,p,q} \implies \nu^{A_v}_{D,p,q}.$$

By Theorem 2.3, this limit differs from $P_{D,p,q} = \nu^{A_h}_{D,p,q}$ because $A_v A_h^{-1} = \mathrm{diag}(a, a^{-1})$ is not a similarity. □

This proves that an appropriate boundary-stability requirement excludes the example. It does not prove that boundary stability plus the three original axioms suffices for classification.

Corollary 5.2 (Boundary stability classifies tangent-dependent affine laws)**.** *Suppose, on smooth bounded Jordan domains with smooth targets, an ensemble has the form* $P_{D,p,q} = \nu_{D,p,q}^{A_{\ell(D,q)}}$*, where* A_ℓ *is an arbitrary orientation-preserving invertible linear map depending only on the unoriented target tangent line. If these laws are continuous under the uniform smooth-boundary approximations used in Proposition 5.1, then one fixed affine map normalizes all of them. If the ensemble is also covariant under all rotations, that fixed map is a similarity.*

Proof. Given any two tangent lines ℓ_1, ℓ_2, take a disk having tangent ℓ_1 at q and apply the preceding local twist with the rotation carrying ℓ_1 to ℓ_2. Every approximating domain uses the constant matrix A_{ℓ_2}. SLE continuity in that fixed metric shows that the limiting law is $\nu_{D,p,q}^{A_{\ell_2}}$, whereas the assumed boundary stability makes it $\nu_{D,p,q}^{A_{\ell_1}}$. Single-pair identifiability implies $A_{\ell_2} A_{\ell_1}^{-1}$ is a similarity. Thus all $A_\ell^{\mathsf{T}} A_\ell$ are proportional, which is exactly one fixed affine family. For rotation covariance its fixed tensor Q satisfies $R^{\mathsf{T}} Q R = Q$ for every rotation R, by the same metric uniqueness, hence Q is scalar. □

5.2 Reversibility eliminates this mechanism inside the affine-germ ansatz

Here is a rigorous limitation of the constructed counterexamples, useful for formulating a more faithful percolation classification. Suppose

$$P_{D,p,q} = A(D,q)_*^{-1} \mathrm{SLE}_6(AD; Ap, Aq)$$

on bounded Jordan domains, and suppose this family is reversible: reversing $P_{D,p,q}$ gives $P_{D,q,p}$. Then, for all p, q,

$$A(D,p)^T A(D,p) = c(D,p,q) A(D,q)^T A(D,q), \qquad c(D,p,q) > 0.$$

Use here the external reversibility theorem of J. Miller and S. Sheffield, *Imaginary geometry III: reversibility of SLE$_\kappa$ for $\kappa \in (4,8)$*, Annals of Mathematics 184 (2016), 455–486, https://annals.math.princeton.edu/2016/184-2/p03. Indeed, ordinary SLE_6 reversibility says that reversing the first curve uses the same matrix $A(D,q)$, now with endpoints q, p. By assumed reversibility this is equal to the law constructed with $A(D,p)$. Single-pair affine identifiability therefore says that $A(D,p)A(D,q)^{-1}$ is a similarity, which is equivalent to the displayed matrix relation. For a rectangle with one endpoint in a horizontal side and the other in a vertical side, the proposed matrices $\mathrm{diag}(1,a)$ and $\mathrm{diag}(a,1)$ violate this relation when $a \neq 1$. Thus the construction explicitly fails reversibility.

This result only eliminates endpoint-dependent anisotropy *within this ansatz*; it does not prove the desired classification after reversibility is added. Near-critical percolation suggests a separate criticality condition remains necessary.

6 Further affine rigidity and target consistency

The following argument concerns ordinary chordal SLE_6; it does not assume conformal invariance of an unknown exploration family. For SLE_6 in $\mathbb{H}$ from 0 to ∞, the left-passage event $L(z)$ is defined by the eventual side of the oriented trace containing z. For a nonsimple trace this is also its eventual Loewner swallowing side. It is invariant under orientation-preserving homeomorphisms of the marked half-plane. A fixed interior point belongs to the trace with probability zero.

Lemma 6.1 (Strict left-passage function)**.** *For $z = x + iy$, $y > 0$,*

$$\mathbb{P}[L(z)] = F(x/y), \qquad F(u) = \frac{\int_{-\infty}^{u}(1+v^2)^{-2/3}\,dv}{\int_{-\infty}^{\infty}(1+v^2)^{-2/3}\,dv}.$$

In particular, F is strictly increasing.

Proof. This is the standard SLE left-passage formula; its short diffusion derivation explains why strict monotonicity holds even for the nonsimple SLE_6 trace. Write $g_t(z) - \sqrt{\kappa}B_t = X_t + iY_t$. Then

$$dX_t = \frac{2X_t}{X_t^2 + Y_t^2}\,dt - \sqrt{\kappa}\,dB_t, \qquad dY_t = -\frac{2Y_t}{X_t^2 + Y_t^2}\,dt.$$

For $U = X/Y$ and $ds = dt/Y_t^2$, Itô's formula gives

$$dU_s = \frac{4U_s}{1+U_s^2}\,ds - \sqrt{\kappa}\,d\widetilde{B}_s.$$

The increasing scale function with limits $0, 1$ solves

$$\frac{\kappa}{2}F'' + \frac{4u}{1+u^2}F' = 0, \qquad F'(u) = C_\kappa(1+u^2)^{-4/\kappa}.$$

The normalizing integral is finite for $\kappa < 8$. The clock runs to infinity up to the point's maximal Loewner lifetime: otherwise $d\log Y_s = -2(1+U_s^2)^{-1}ds$ keeps Y positive and the diffusion extends. The bounded martingale $F(U_s)$ converges. A limit in $(0,1)$ would force U_s to converge to a finite value and its martingale quadratic variation to be infinite, a contradiction. Thus $U_s \to -\infty$ or $+\infty$. Optional stopping and the usual topological side identification for chordal Loewner chains give the formula at $\kappa = 6$. □

Proposition 6.2 (Linear symmetries of half-plane SLE_6)**.** *If an invertible real-linear map M carries $SLE_6(\mathbb{H}, 0, \infty)$ to $SLE_6(M\mathbb{H}, 0, \infty)$, then M is a Euclidean similarity, possibly reversing orientation.*

Proof. Reflection invariance reduces to $\det M > 0$. Rotate $M\mathbb{H}$ back to $\mathbb{H}$. The resulting map is $T(x,y) = (ax + by, cy)$ with $a, c > 0$. By topological preservation of the side event and Lemma 6.1,

$$F(u) = F((au+b)/c) \qquad (u \in \mathbb{R}).$$

Strict monotonicity implies $a = c$, $b = 0$. Undo the rotation. □

6.1 Target consistency is a substantive additional condition

Additional axiom; not a consequence proved here. For two targets q_1, q_2 on a boundary arc J whose closure avoids p, require explorations towards them to couple identically until first hitting J (or use the usual stopping formulation before target separation). A common percolation coloring gives this: the two boundary conditions disagree only on J, so the exploration algorithms see identical colors until reaching it. The stated locality axiom fixes q and does not assert this property.

Proposition 6.3 (Target consistency removes variable affine target metrics)**.** *Suppose D is a bounded Jordan domain analytic near p, and*

$$\mathcal{P}_{D,p,q} = (A_q^{-1})_*\,\mathrm{SLE}_6(A_qD, A_qp, A_qq).$$

If target consistency holds for q_1, q_2 on an arc avoiding p, then $A_{q_2}A_{q_1}^{-1}$ is a similarity. In particular, their metric tensors $A_q^{\mathsf{T}}A_q$ are proportional.

Proof. Translate p to zero and dilate by $r \to \infty$. The domains converge locally to the tangent half-plane H_p, and targets tend to infinity. For each fixed q, conformal kernel convergence at the analytic start and the usual initial-segment construction of SLE give limiting law

$$(A_q^{-1})_* \operatorname{SLE}_6(A_q H_p, 0, \infty).$$

Convergence can be tested on paths stopped on crosscuts strictly inside fixed half-disks, followed by exhaustion. The dilated disagreement arc escapes every fixed neighborhood, so target consistency says that the two limiting half-plane laws coincide. Apply A_{q_1}, rotate the image half-plane, and use Proposition 6.2. □

This eliminates the specific target-tangent obstruction, not every possible counterexample. The proof uses initial-segment convergence for ordinary SLE_6, not for an unknown general ensemble.

6.2 A compatibility graph for all target-germ affine decorations

The previous target-consistency proposition also gives a useful classification within the entire class of affine decorations, without a claim about arbitrary exploration ensembles. Define a graph whose vertices are admissible embedded target germs, modulo translations, and join G_1, G_2 when they can both occur on one bounded Jordan domain with an analytic starting point p elsewhere. The targets must lie on a boundary arc whose closure avoids p. Assign to each vertex the projective positive-definite tensor $[Q(G)] = [A(G)^{\mathsf{T}} A(G)]$.

Corollary 6.4 (Target-germ metrics are constant on compatibility components)**.** *For a target-germ affine SLE_6 decoration satisfying target consistency, $[Q(G)]$ is constant on every connected component of this graph. If the graph is connected, the decoration is one fixed-affine SLE_6 family.*

Proof. Proposition 6.3 makes the tensors proportional across every edge. Propagate along finite paths. If a component has representative matrix A_0, proportionality implies $A(G) = c_G R_G A_0$ with $c_G > 0$ and R_G orthogonal. Similarity covariance of SLE_6 then gives

$$A(G)_*^{-1} \operatorname{SLE}_6(A(G)D, A(G)p, A(G)q) = (A_0^{-1})_* \operatorname{SLE}_6(A_0 D, A_0 p, A_0 q),$$

independently of G. □

For ordinary smooth local boundary germs, connectivity is suggested by a simple planar surgery: place two small representatives far apart, retain their marked local boundary arcs, join the remaining endpoints with disjoint arcs to form one Jordan boundary, and place an analytic start on a joining arc. A fully general surgery statement for nonlocally connected prime-end impressions is not asserted here; the graph formulation keeps this issue explicit. For the straight-line tangent germs in the counterexample, joint realization on an ordinary rounded rectangle already connects horizontal and vertical targets and excludes their distinct tensors.

6.3 Finite rotations and a fixed anisotropy

Proposition 6.5. *Suppose an ensemble has one fixed affine SLE_6 normalization with linear part A, and is covariant under rotation R_θ, where $\theta \notin \pi\mathbb{Z}$. Then $A^{\mathsf{T}}A$ is scalar, so A is a similarity. A half-turn alone does not suffice.*

Proof. The map $AR_\theta A^{-1}$ preserves the SLE_6 family and hence is a similarity. Its determinant is 1, so it is orthogonal. For $Q = A^{\mathsf{T}}A$ this says $R_\theta^{\mathsf{T}} Q R_\theta = Q$. A symmetric real 2×2 matrix commuting with a rotation other than 0 or π is scalar. For $R_\pi = -I$, every Q satisfies the relation. □

The fixed-normalizer hypothesis is essential: the target-tangent construction can be covariant under every rotation and still have no fixed normalizer. Rotation symmetry by itself does not repair the conjecture.

7 What locality actually says about a semimartingale driving process

Status. The coordinate-change identities below are proved, under explicitly stated continuous-semimartingale assumptions. The conditional martingale theorem does not require absolute continuity of quadratic variation. They give a conditional SLE classification. The original three axioms do not currently imply these semimartingale or local-martingale assumptions, and no such implication is claimed. In particular the domain Markov property does not make a driving process Markov in its single real coordinate.

7.1 Deletion of a hull and the drift anomaly

Use chordal Loewner normalization

$$\partial_t g_t(z) = \frac{2}{g_t(z) - W_t}, \qquad g_t(z) = z + \frac{2t}{z} + O(z^{-2}).$$

Suppose on an interval before collision with a fixed hull that

$$dW_t = dM_t + b_t\,dt, \qquad d\langle M\rangle_t = v_t\,dt, \tag{L1}$$

where M is a continuous local martingale and $v_t \geq 0$. Let A be a bounded half-plane hull away from the initial tip, and let $h_0 : \mathbb{H} \setminus A \to \mathbb{H}$ have hydrodynamic normalization at infinity. Grow the same trace in the deleted domain, mapping by h_0 first, and denote its Loewner maps by $\widehat{g}_u$. Up to the first encounter with A, write

$$h_t = \widehat{g}_{u(t)} \circ h_0 \circ g_t^{-1}, \qquad \widehat{W}_{u(t)} = h_t(W_t), \qquad \dot{u}(t) = h_t'(W_t)^2. \tag{L2}$$

The map h_t is analytic across the real line near W_t, by Schwarz reflection, and $h_t'(W_t) > 0$. These formulas are local near the driving point; no differentiability of the trace is used.

Differentiating the composition in (L2) gives

$$\partial_t h_t(z) = \frac{2h_t'(W_t)^2}{h_t(z) - h_t(W_t)} - \frac{2h_t'(z)}{z - W_t}. \tag{L3}$$

For $z = W_t + \epsilon$ the first term is $2h_t'(W_t)/\epsilon - h_t''(W_t) + O(\epsilon)$, while the second term is $2h_t'(W_t)/\epsilon + 2h_t''(W_t) + O(\epsilon)$. Consequently

$$(\partial_t h_t)(W_t) = -3h_t''(W_t). \tag{L4}$$

The random field $h_t(z)$ has finite variation in time locally before collision, so ordinary Itô's formula (equivalently the finite-variation case of Itô–Wentzell) applies to $h_t(W_t)$. Thus

$$d\widehat{W}_{u(t)} = h_t'(W_t)\,dM_t + \left[h_t'(W_t)b_t + \left(\frac{v_t}{2} - 3\right) h_t''(W_t)\right] dt. \tag{L5}$$

After the capacity time change, its characteristics are therefore

$$\boxed{\widehat{v}_{u(t)} = v_t, \qquad \widehat{b}_{u(t)} = \frac{b_t}{h_t'(W_t)} + \left(\frac{v_t}{2} - 3\right) \frac{h_t''(W_t)}{h_t'(W_t)^2}.} \tag{L6}$$

The coefficient -3, and hence the exceptional variance 6, follow from the normalization $\partial_t g_t = 2/(g_t - W_t)$. A different capacity normalization changes both constants consistently.

Logical content. Locality identifies the law of the transformed driving process with the law prescribed for the deleted physical domain in its own conformal chart. It does *not* identify that law with the original driving law in its original chart. In particular the two drifts in (L6) need not vanish, or be equal. A geometry-dependent drift can transform with the second term in (L6); deleting that term without an additional hypothesis is exactly where an invalid proof of $v = 6$ would enter. Physical translation invariance acts by translating the physical domain and trace. It supplies no invariance under the nonlinear map h_t and cannot by itself justify identifying these two characteristics.

7.2 A precise conditional classification by driving martingales

Conditional theorem (proved).
Suppose an ensemble of continuous Loewner-generating chordal traces satisfies strong domain Markov and locality, with the conventional endpoint-preserving hull deletions. Suppose also that, for every deterministic marked domain, its capacity driving process in a chordal conformal chart is a continuous local martingale in its natural filtration. Assume the usual measurability and restart consistency of this family, so that the strong domain Markov property applies in the chart at a random slit domain. Then

$$W_t - W_0 = \sqrt{6}\, B_t,$$

and the ensemble is ordinary chordal SLE_6. No absolute continuity or differentiability assumption on quadratic variation is needed. The chart premise is independent of which chordal chart is selected, since two such charts differ by $z \mapsto az + b$, $a > 0$, and the corresponding deterministic capacity time change preserves local martingales.

Proof. Fix an initial marked domain and map it to $(\mathbb{H}, 0, \infty)$. Delete the inverse image of a vertical slit at $x \neq 0$, of height $2\sqrt{s}$ for $s > 0$. The hydrodynamic map of this slit is

$$h_0(z) = x + \sqrt{(z-x)^2 + 4s}, \tag{L7}$$

with the branch asymptotic to $z - x$ at infinity. At zero,

$$h_0'(0) = \frac{|x|}{\sqrt{x^2+4s}}, \qquad h_0''(0) = -\frac{4s\ \mathrm{sgn}(x)}{(x^2+4s)^{3/2}} \neq 0. \tag{L8}$$

Let $C_t = \langle W \rangle_t$, initially with no regularity beyond the continuity and monotonicity of quadratic variation. The general Itô version of (L5) is

$$d\widehat{W}_{u(t)} = h_t'(W_t)\, dW_t + \frac{1}{2} h_t''(W_t)\, dC_t - 3h_t''(W_t)\, dt. \tag{L8a}$$

By locality and the hypothesis on the deleted domain, $\widehat{W}_u$ is a continuous local martingale; its time change $\widehat{W}_{u(t)}$ is a local martingale in the same trace filtration. Indeed, the inverse clock $t(u)$ is the original-domain capacity of the initial trace reconstructed from the deleted-domain curve. It is an adapted, continuous increasing functional of that trace, so $u(t)$ is a stopping time in the deleted-driver filtration. The usual stopping-time change theorem, after localization, therefore applies. That filtration is also the one generated by W_t, since each driving function determines its initial hull and trace. Localizing before collision and before the chart derivatives become unbounded, the stochastic integral in (L8a) is a local martingale. Its remaining continuous finite-variation term must therefore vanish identically. As signed measures,

$$h_t''(W_t)\big(dC_t - 6\, dt\big) = 0. \tag{L8b}$$

The second derivative starts nonzero by (L8) and remains so on a positive, possibly random, initial time interval. Equation (L8b) proves $C_t = 6t$ on that interval.

This conclusion applies at the start of every deterministic marked domain. To propagate it without any density assumption, set

$$T = \inf\{t \geq 0 : C_t - 6t \neq 0\}, \qquad \inf \varnothing = \infty.$$

In the usual right-continuous filtration this is a stopping time: the first entrance time of the continuous adapted process $C_t - 6t$ into the open set $\mathbb{R} \setminus \{0\}$. The positive initial zero interval shows $T > 0$ almost surely. If $T < \infty$, continuity implies $C_T - 6T = 0$, and by the definition of T there must be nonzero values arbitrarily soon after T. Restarting the exploration at T using the strong domain Markov property gives a positive interval on which $C_{T+s} - C_T = 6s$, a contradiction. (One may localize T before deterministic bounds and before any terminal time.) Hence $C_t = 6t$ throughout. Lévy's characterization now gives the asserted Brownian driving law, and the Loewner correspondence identifies the trace as chordal SLE_6. □

Translation invariance was not needed in this conditional theorem. The additional condition is substantial: it asks for local martingales *in the conformal charts of all physical domains*, not just one chosen half-plane. A possible route toward universality would be to derive such a martingale condition from suitable physical observables. Nothing in this calculation constructs those observables.

7.3 A constant-characteristic version and an explicit two-slit test

A related sufficient assumption is that every marked-domain driving process has the same deterministic constant variance κ and drift b, in its capacity chart. Under locality (L6), a deletion with initial derivatives $d = h_0'(0)$ and $e = h_0''(0)$ must obey

$$b = \frac{b}{d} + \left(\frac{\kappa}{2} - 3\right)\frac{e}{d^2}. \tag{L9}$$

Choose the two vertical slits in (L7) at x and $-x$ with the same height. Their derivatives have the same $d \in (0, 1)$ and opposite nonzero e. Subtracting their instances of (L9) yields $\kappa = 6$; either equation then gives $b = b/d$, hence $b = 0$. Thus the only constant-characteristic ensemble in this class is SLE_6.

This two-slit computation is a useful check on the signs and constants, but its universality premise is much stronger than physical translation invariance. The current bottleneck is to prove an adequate chart-martingale or characteristic condition from genuinely coarse properties of critical percolation, rather than to manipulate (L6) further.

8 Dilation symmetry and criticality

8.1 Dilation covariance and subsequential limits

Lemma 8.1. *Every family with a fixed affine SLE_6 normalization is covariant under all scalar dilations: $\mathcal{P}_{rD,rp,rq} = r_*\mathcal{P}_{D,p,q}$ for $r > 0$.*

Proof. Dilations commute with the linear part of the normalizer; conjugating by its full affine map gives a Euclidean similarity. Use SLE similarity covariance. □

Lemma 8.2 (The cluster set, not each cluster point, inherits dilations)**.** *Let X be a topological space of ensembles with continuous dilation actions T_r. Suppose $P_\delta \in X$ has exact mesh covariance $T_r P_\delta = P_{r\delta}$ and is relatively compact as $\delta \downarrow 0$. Its cluster set $\mathcal{C}$ satisfies $T_r\mathcal{C} = \mathcal{C}$. If $\mathcal{C} = \{P\}$, then $T_r P = P$ for every $r > 0$.*

Proof. If $P_{\delta_n} \to P$, continuity gives $P_{r\delta_n} = T_r P_{\delta_n} \to T_r P$. Thus $T_r\mathcal{C} \subset \mathcal{C}$; use $T_{1/r}$ for equality. □

For laws indexed by marked domains the action is $(T_r P)_{D,p,q} = r_* P_{r^{-1}D, r^{-1}p, r^{-1}q}$. One must choose a topology in which the actions are continuous; this is not silently supplied by abstract tightness. The distinction is real: on $X = S^1$, take $P_\delta = \exp(i\omega \log \delta)$ and $T_r z = \exp(i\omega \log r) z$, $\omega \neq 0$. The cluster set is all of S^1 but has no member fixed by every dilation. Uniform nondegenerate observable bounds do not remove this possibility: $\frac{1}{2} + \varepsilon \sin(\omega \log(R/\delta))$, $0 < \varepsilon < 1/2$, remains uniformly bounded while oscillating in scale phase. This is a topological example, not a percolation counterexample. It nevertheless explains why deriving dilation invariance of every subsequential limit from tightness or RSW alone would be circular.

8.2 All-scale RSW and a nonzero near-critical parameter

Proposition 8.3 (A finite correlation scale conflicts with all-scale RSW)**.** *Suppose continuum colorings P^λ obey $T_r P^\lambda = P^{r^{-\alpha}\lambda}$ for some $\alpha > 0$. For a fixed rectangle shape let $c(\lambda)$ be its unit-scale white crossing probability. Assume $c(\lambda) \to 1$ as $\lambda \to +\infty$ and $c(\lambda) \to 0$ as $\lambda \to -\infty$. If one P^λ has white crossing probabilities in $[\varepsilon, 1-\varepsilon]$ at every scale $R > 0$, with fixed $\varepsilon > 0$, then $\lambda = 0$.*

Proof. Scaling a size-R rectangle down to unit size changes the parameter to $R^\alpha \lambda$. Its crossing probability is $c(R^\alpha \lambda)$, which tends to 1 or 0 for positive or negative λ. □

This is a conditional calculation, separate from existence and regularity of near-critical continuum percolation. For standard triangular-lattice near-critical percolation $\alpha = 3/4$. RSW merely on each bounded scale range, with constants depending on that range, does not exclude nonzero mass. Moreover the target-tangent construction has bounded anisotropy and no preferred scale; removing a near-critical parameter does not address the target-consistency obstruction.

9 Near-critical obstruction: what is rigorous and what is not yet transferred

The missing criticality assumption is a separate obstruction from terminal germ dependence. On triangular mesh η, write

$$r(\eta) = \frac{\eta^2}{\alpha_4^\eta(\eta, 1)}, \qquad p_\eta = \frac{1}{2} + \lambda r(\eta), \qquad \lambda \neq 0,$$

where α_4 is the critical alternating four-arm probability. The normalization is $r(\eta) = \eta^{3/4+o(1)}$. At every mesh the exploration has exact spatial domain Markov and locality properties for every p_η. These two properties alone cannot express criticality.

Established external facts (not reproved here). Nolin–Werner, *Asymmetry of near-critical percolation interfaces*, JAMS 22 (2009), 797–819, https://arxiv.org/abs/0710.1470, establish subsequential limits of near-critical interfaces and their singularity with respect to ordinary SLE_6. Garban–Pete–Schramm, *The scaling limits of near-critical and dynamical percolation*, JEMS 20

(2018), 1195–1268, https://arxiv.org/abs/1305.5526, construct the configuration-valued near-critical scaling limit and establish its conformal covariance. Their Section 13.2 explicitly distinguishes the measurable trace of an interface from its order as a curve: the latter uniqueness is not proved there. Their Section 11 Markov property concerns change of the near-critical parameter, not the spatial exploration stopping-time property needed here. Thus neither statement may silently be substituted for full convergence of ordered exploration kernels in random slit domains. A 2025 Durham thesis of Léonie Papon, introduction, https://etheses.durham.ac.uk/id/eprint/16026/1/Papon-thesis-v2.pdf, also records ordered near-critical percolation interface uniqueness as a remaining challenge. These observations are limits of the sources checked, not a proof that no later theorem exists.

A simple necessary test for fixed affine SLE. Every fixed affine image of the ordinary SLE ensemble is covariant under all scalar dilations. Indeed, if the linear part of the affine map is L, then $L(rx) = rLx$; the translation part is absorbed by translation covariance. Consequently a genuine failure of dilation covariance excludes *every* fixed affine normalization. Singularity relative to ordinary SLE alone does not have this implication: an anisotropic affine image may also be singular.

Under a physical dilation by $s > 0$, the near-critical parameter changes as

$$s\,\omega^{\lambda} \overset{\text{law}}{=} \omega^{s^{-3/4}\lambda}.$$

The sign of the exponent follows already on the lattice: the original mesh η becomes $s\eta$ while the same occupation probability has effective parameter $\lambda r(\eta)/r(s\eta) \to s^{-3/4}\lambda$. Known nonconstant crossing probabilities as functions of λ therefore witness a macroscopic scale in the configuration law. Transferring this witness to a full domain-indexed curve ensemble requires the ordered-kernel step just identified. This investigation therefore records near-critical interfaces as a *conditional additional counterexample*, not as the proof of the counterexample in the main text.

9.1 An explicit limit-transfer lemma

Here is a precise sufficient condition for the commonly asserted passage of the spatial Markov property to scaling limits. It isolates the missing assumption rather than labeling it automatic.

Let (I_n, D_n, Y_n) denote respectively the explored initial segment, remaining marked domain, and future curve at a stopping time. Suppose these random objects take values in Polish spaces and converge jointly in distribution to (I, D, Y). Let K_n be the future-curve kernels, so

$$\mathbb{E}[f(Y_n) \mid I_n, D_n] = K_n f(D_n).$$

Assume there is a Borel probability kernel K such that, for every bounded continuous f and every sequence $d_n \to d$ with d in a full-law set for D, one has $K_n f(d_n) \to K f(d)$. Then

$$\mathbb{E}[f(Y) \mid I, D] = K f(D).$$

Proof. By a Skorokhod representation the triples may be assumed to converge almost surely. For any bounded continuous test function h of (I, D), the exact finite-n identity gives

$$\mathbb{E}[h(I_n, D_n) f(Y_n)] = \mathbb{E}[h(I_n, D_n) K_n f(D_n)].$$

Bounded convergence and the sequential-kernel hypothesis give the same identity at the limit, with $Kf(D)$ on the right. The monotone class theorem extends the identity to bounded measurable h, and a countable convergence-determining class of f identifies the conditional kernel. □

The decisive hypothesis controls kernels at *random slit domains*; convergence for one deterministic Jordan domain is insufficient. Analogous coupled convergence can transfer locality before first approach to a removed closed set: stop first at positive-distance neighborhoods, where uniform curve convergence leaves the two coupled initial segments identical, then decrease the distance to zero. Exact first-hit stopping needs a separate continuity argument and is not implicit in this observation.

10 Affine identifiability without Jordan boundary regularity

Status and interpretation. The extension proved here uses only standard quasiconformal compactness, prime-end extension for quasiconformal disk maps, Fatou's theorem for bounded analytic functions, and uniqueness of bounded harmonic functions from their almost-everywhere disk boundary values. It removes the Jordan assumption from affine metric identifiability. At completely arbitrary prime ends a conformal image of SLE need not have a Euclidean-continuous endpoint; its law is therefore understood in the prime-end compactification, or as its conformal hull exploration. The result applies in particular whenever the physical curves are continuous, as in the exploration setting of interest. No classification of arbitrary exploration ensembles is asserted.

10.1 A quasiconformal disk map fixing the boundary stays close to the identity

Lemma (proved). For each finite $K \geq 1$ there exists $C(K) < \infty$ such that every orientation-preserving K-quasiconformal self-homeomorphism $h : \mathbb{D} \to \mathbb{D}$ whose boundary extension fixes every point of $\partial\mathbb{D}$ satisfies

$$d_{\mathbb{D}}(h(z), z) \leq C(K) \qquad (z \in \mathbb{D}). \tag{RA1}$$

Here $d_{\mathbb{D}}$ is hyperbolic distance.

Proof. First there is $r(K) < 1$ with $|h(0)| \leq r(K)$ for all such maps. To see this, extend h to the sphere by reflection in the unit circle:

$$\widetilde{h}(z) = \frac{1}{\overline{h(1/\bar{z})}} \quad (|z| > 1).$$

This is a K-quasiconformal sphere map, and fixes three prescribed distinct points of the unit circle. The standard compactness theorem for normalized K-quasiconformal sphere maps, including compactness of their inverses, precludes a sequence with $h_n(0) \to \zeta \in \partial\mathbb{D}$: a limiting quasiconformal homeomorphism would send both 0 and the fixed boundary point ζ to ζ. Thus such an $r(K)$ exists. For an arbitrary z choose a disk automorphism a_z sending z to 0. The map $a_z \circ h \circ a_z^{-1}$ is again K-quasiconformal with identity boundary map, so

$$|a_z(h(z))| \leq r(K).$$

Conformal invariance of the hyperbolic metric gives (RA1), for example with $C(K) = 2 \operatorname{arctanh} r(K)$ in the usual normalization. □

Consequence (proved). For every $\zeta \in \partial\mathbb{D}$, the image of the radial approach $r\zeta$ under h is a nontangential approach to the same ζ, with aperture bounded solely in terms of K. Indeed rotate so $\zeta = 1$ and write

$$h(r) = \frac{r + u_r}{1 + r u_r}, \qquad |u_r| \leq r(K).$$

Then

$$|1 - h(r)| \le (1 - r)\frac{1 + r(K)}{1 - r(K)},$$

while

$$1 - |h(r)| \ge \frac{1 - r}{2}\frac{1 - r(K)}{1 + r(K)}.$$

The ratio $|1 - h(r)|/(1 - |h(r)|)$ is therefore uniformly bounded. In particular $h(r\zeta) \to \zeta$ within a fixed Stolz region.

10.2 The identifiability theorem on any bounded simply connected domain

Theorem (proved). Let D be any bounded simply connected planar domain and let p, q be distinct prime ends. For orientation-preserving invertible real-linear maps A, B, define

$$\nu^A_{D,p,q} = (A^{-1})_* \,\mathrm{Law}(\mathrm{SLE}_6(AD; Ap, Aq)).$$

Then

$$\nu^A_{D,p,q} = \nu^B_{D,p,q} \quad \Longleftrightarrow \quad BA^{-1} \text{ is a Euclidean similarity} \quad \Longleftrightarrow \quad B^{\mathsf{T}}B = cA^{\mathsf{T}}A \quad (c > 0). \tag{RA2}$$

The statement holds for exploration laws in the prime-end compactification, and hence for their physical curves whenever defined continuously.

Proof. Set $M = BA^{-1} : AD \to BD$. This bilipschitz map carries null crosscut chains to null crosscut chains, with equivalent chains remaining equivalent; thus it induces the natural homeomorphism of prime-end compactifications. Choose conformal charts $\phi_A : AD \to \mathbb{H}$ and $\phi_B : BD \to \mathbb{H}$ sending the corresponding source and target to $0, \infty$. The interior map

$$H = \phi_B \circ M \circ \phi_A^{-1} : \mathbb{H} \to \mathbb{H}$$

is quasiconformal with a boundary extension to $\mathbb{R} \cup \{\infty\}$, fixing $0, \infty$. The boundary map is increasing. The boundary-swallowing race of the main record remains a prime-end topological event: an affine homeomorphism preserves the order in which boundary ends are cut off from the target by the explored filled hull. Equality of the two path laws, followed by strictness of that race probability, therefore gives, for every $x < 0 < y$,

$$\frac{-H(x)}{H(y) - H(x)} = \frac{-x}{y - x}.$$

Consequently $H(x) = cx$ on the entire extended real boundary, for some $c > 0$. This step uses prime-end boundary values of H, not Euclidean boundary values of either Riemann map.

Pass to disk charts, absorbing the conformal map $z \mapsto cz$ into the second chart. We obtain bounded conformal maps

$$\psi_A : \mathbb{D} \to AD, \qquad \psi_B : \mathbb{D} \to BD$$

such that

$$h = \psi_B^{-1} \circ M \circ \psi_A$$

is a K-quasiconformal disk self-homeomorphism with identity boundary map. Thus pointwise in the disk,

$$M\psi_A = \psi_B \circ h. \tag{RA3}$$

Fatou's theorem gives radial limits $\psi_A^*(\zeta)$ and nontangential limits $\psi_B^*(\zeta)$ for almost every circle point ζ. The preceding lemma shows that $h(r\zeta)$ approaches that same ζ nontangentially. Taking $r \uparrow 1$ in (RA3) yields

$$M\psi_A^*(\zeta) = \psi_B^*(\zeta) \quad \text{for almost every } \zeta \in \partial\mathbb{D}. \tag{RA4}$$

Both real coordinates of $M\psi_A$ are bounded harmonic functions on the disk, since M is real-linear and ψ_A is bounded analytic. The coordinates of ψ_B are also bounded harmonic. Uniqueness of the Poisson representation for bounded harmonic functions now gives

$$M\psi_A = \psi_B \qquad \text{throughout } \mathbb{D}.$$

Thus M is conformal on the nonempty open set AD. Writing $M(z) = \alpha z + \beta\bar{z}$, holomorphicity of $M\psi_A$ gives $\beta\overline{\psi_A'(z)} = 0$. Since ψ_A' never vanishes, $\beta = 0$. Invertibility and orientation imply that M is a direct Euclidean similarity. The converse is SLE similarity covariance. Reflections reduce orientation-reversing normalizers to this case. □

The boundedness assumption was used exactly when invoking bounded analytic Fatou limits and bounded harmonic uniqueness. No assertion about arbitrary unbounded domains follows without a growth or uniqueness condition.

10.3 A necessary and sufficient DMP condition inside the affine ansatz

Suppose, as an explicit additional ansatz, that an ensemble has the form

$$P_{D,p,q} = \nu_{D,p,q}^{A(D,p,q)} \tag{RA5}$$

for a deterministic measurable affine label on each marked bounded simply connected domain. Remove the inessential positive scalar from its metric by putting

$$T(D,p,q) = \frac{A(D,p,q)^{\mathsf{T}} A(D,p,q)}{\sqrt{\det(A(D,p,q)^{\mathsf{T}} A(D,p,q))}}. \tag{RA6}$$

Thus T is a positive-definite symmetric matrix of determinant one. Assume the usual Borel family and strong conditional-kernel formulation, so that equality of conditional path laws at a stopping time is an equality of probability kernels almost surely.

Corollary (proved). Within ansatz (RA5), the strong domain Markov property holds if and only if, for every starting marked domain and every admissible stopping time τ,

$$T(D_\tau, p_\tau, q) = T(D,p,q) \qquad \text{almost surely.} \tag{RA7}$$

Here D_τ is the target component of the remaining domain. It need not have Jordan boundary.

Proof. Fix the starting label $A = A(D,p,q)$. Under the fixed-A affine SLE law, ordinary SLE's strong domain Markov property gives, conditional on the explored initial segment, the remaining law

$$\nu_{D_\tau,p_\tau,q}^{A}.$$

Strong DMP for (RA5) demands that this be $\nu_{D_\tau,p_\tau,q}^{A(D_\tau,p_\tau,q)}$ almost surely. The rough-domain theorem (RA2), applied at each realized slit domain, is equivalent to equality of their normalized metric tensors, namely (RA7). Conversely (RA7) and similarity covariance identify these two conditional laws, proving strong DMP. □

This corollary classifies the DMP constraint only *within the stated ansatz*: the affine tensor must be a conserved quantity of the exploration state. A target-germ tensor supplies exactly such a conserved quantity. The corollary neither proves that a general ensemble has affine-SLE form nor that every conserved tensor must arise from a target germ. Locality, translation covariance, target consistency, and boundary continuity impose further constraints which are not characterized by (RA7) alone.

11 Local target consistency passes to marginal curve limits

Scope. This lemma transfers an exploration coupling only inside a fixed neighborhood of the common start. It needs convergence of the two marginal curve laws, but neither convergence of random slit-domain kernels nor a limiting strong domain Markov theorem. It does not assert a coupling of the limiting curves all the way to a distant boundary arc.

11.1 Stopping an ordered unparameterized curve on a circle

Fix $p \in \mathbb{R}^2$ and $R_0 > 0$. Work in the usual metric space of oriented continuous curves modulo increasing reparameterization, with the uniform Fréchet topology. Curves start at p and end outside $\overline{B(p, R_0)}$. Small changes of the starting points are harmless; for the sequence version they need only lie inside every fixed smaller ball for sufficiently large indices. For a continuous representative $\gamma : [0, 1] \to \mathbb{R}^2$ and $0 < r < R_0$, put

$$\tau_\gamma(r) = \inf\{t : |\gamma(t) - p| \geq r\}, \qquad E_r\gamma = \gamma|_{[0,\tau_\gamma(r)]}. \tag{LT1}$$

The stopped path is regarded as an ordered unparameterized curve, so this definition is independent of pauses or changes of clock.

Lemma 11.1 (Almost every circle is a continuity circle)**.** *For each fixed continuous curve γ, the map $(\eta, s) \mapsto E_s\eta$ is continuous at (γ, r) for all but at most countably many radii $r \in (0, R_0)$. The stopping map is Borel. Consequently, for any probability law μ on these curves, E_r is μ-almost surely continuous for Lebesgue-almost every deterministic r.*

Proof. Choose any continuous representative of γ. The function $r \mapsto \tau_\gamma(r)$ is nondecreasing and hence has at most countably many discontinuities. Suppose it is continuous at r. If $(\gamma_n, r_n) \to (\gamma, r)$ in the product of Fréchet and ordinary metrics, choose increasing reparameterizations for which representatives $\widetilde{\gamma}_n$ converge uniformly to γ. Writing $\varepsilon_n = \|\widetilde{\gamma}_n - \gamma\|_\infty + |r_n - r|$, first-hit times obey, for every slightly larger positive error tending to zero,

$$\tau_\gamma(r - 2\varepsilon_n) \;\leq\; \tau_{\widetilde{\gamma}_n}(r_n) \;\leq\; \tau_\gamma(r + 2\varepsilon_n). \tag{LT2}$$

If $\varepsilon_n = 0$ the assertion is immediate; using a strictly larger error avoids any immaterial equality convention. Continuity of τ_γ at r gives convergence of the middle times to $\tau_\gamma(r)$. Uniform continuity of γ then makes the stopped representatives, linearly recoded onto $[0, 1]$, converge uniformly. This proves joint continuity at (γ, r).

The stopping map is understood on all curves reaching the selected radius, including already stopped curves reaching a larger radius. For parameterized curves the stopping time is Borel, since

$$\{\tau_\gamma(r) \leq t\} = \left\{ \max_{0 \leq u \leq t} |\gamma(u) - p| \geq r \right\}.$$

Stopping and linear recoding are therefore Borel path operations. They commute with reparameterization and give the usual Borel stopping map on ordered unparameterized curves.

For completeness, the almost-every-radius assertion does not require a measurable choice of representatives. For a map between metric spaces its set of continuity points is a G_δ set, obtained from its oscillations on neighborhoods. Apply this to the joint stopping map. Its discontinuity set in the product of curve space and radius space is Borel, and every vertical section has Lebesgue measure zero by the countability argument above. Fubini's theorem now gives the assertion under μ. □

Why a bad radius can occur. A path may first touch a circle, then spend a nontrivial segment inside it, and only later cross it. A small outward perturbation can postpone its first exit until that later crossing. This is why stopping continuity cannot simply be claimed at every deterministic radius for every path. The preceding lemma handles these tangencies without excluding them by an unproved regularity assumption.

11.2 Transfer using only two marginal limits

Theorem 11.2 (Transfer of local target consistency)**.** *For $i = 1, 2$, let γ_n^i be oriented continuous random curves whose marginal laws converge weakly in the uniform Fréchet topology to laws μ^i. Assume the limiting curves start at p and end outside $\overline{B(p, R_0)}$. Suppose that for every fixed $r \in (0, R_0)$ and all sufficiently large n,*

$$\mathrm{Law}(E_r\gamma_n^1) = \mathrm{Law}(E_r\gamma_n^2). \tag{LT3}$$

Then the same equality holds for the limiting curves at every *$r \in (0, R_0)$:*

$$(E_r)_*\mu^1 = (E_r)_*\mu^2. \tag{LT4}$$

The hypothesis (LT3) may equivalently be replaced by a discrepancy tending to zero in any metric inducing weak convergence of the stopped laws.

Proof. Lemma 11.1 gives a full-Lebesgue-measure set of radii at which E_r is almost surely continuous under both limiting laws. At each such radius the continuous mapping theorem and (LT3) give (LT4). This part needs no joint convergence of (γ_n^1, γ_n^2).

To obtain every smaller radius, fix arbitrary $r < R_0$ and choose a good $s \in (r, R_0)$. Equality of the stopped-at-s laws remains equality after applying the Borel map E_r. Since $E_r(E_s\gamma) = E_r\gamma$ for $r < s$, this proves (LT4) at r without requiring any continuity of E_r there. □

11.3 Application to two percolation targets

Let D be a Jordan domain, p a common boundary start, and let the targets q_1, q_2 lie on a boundary arc J with $p \notin \overline{J}$. The boundary colors for the two Dobrushin explorations differ only on the arc between the targets. Use identical independent bulk colors and identical boundary colors outside a lattice approximation J_n of this disagreement arc, with a common discrete starting edge. The discrete exploration rules produce exactly the same ordered initial path until the exploration first encounters J_n. Choose

$$0 < R_0 < \mathrm{dist}(p, \overline{J}).$$

For all large n, the approximating disagreement arc stays outside the closed ball $\overline{B(p, R_0)}$. Thus (LT3) holds inside every smaller ball. If a common subsequence makes each marginal exploration

converge, Theorem 11.2 supplies local target consistency for their limiting initial segments. The lattice approximations must in fact have the stated common-start/common-color coupling; a small displacement of two unrelated explorations alone would not justify pathwise closeness.

Consequence for affine decorations. The affine-metric elimination argument needs only equality of these local prefix laws, not equality until the full distant arc. Suppose D is analytic near p and the two limiting marginals have laws

$$(A_i^{-1})_* \operatorname{SLE}_6(A_i D, A_i p, A_i q_i), \qquad i = 1, 2.$$

Translate p to zero and blow up space. By conformal kernel convergence at the smooth start, the local initial laws tend to

$$(A_i^{-1})_* \operatorname{SLE}_6(A_i H_p, 0, \infty),$$

where H_p is the tangent half-plane. Equality in every sufficiently small physical ball transfers after dilation to equality of the limiting stopped laws in every fixed half-disk (first at common continuity radii, then all smaller radii by the same composition argument). These local restrictions determine the entire half-plane law. Strict left-passage identifiability therefore makes $A_2 A_1^{-1}$ a similarity.

Consequently the target-tangent affine counterexample cannot be the family of simultaneous two-target subsequential percolation limits on a fixed smooth domain with targets carrying different tangent directions. This exclusion is independent of the more difficult transfer of a strong domain Markov property to random limiting slit domains.

12 Affine metrics encoded in initial path germs

Status and motivation. The result below is proved for the *oriented initial trace germ* of an affine image of half-plane SLE_6 at its smooth starting boundary point. It does not construct a source-dependent domain Markov family. In particular, it does not establish that the metric can be recovered from the unlabelled boundary germ at a later random exploration tip. Those distinctions are important when assessing whether target consistency alone could repair the classification conjecture.

12.1 State space and the meaning of a trace germ

We retain the chronological order of a path but forget its original clock. For the half-plane curves used here, the usual half-plane-capacity clock provides a canonical representative: the filled hull at time t has hydrodynamic map $g_t(z) = z + 2t/z + o(1/z)$. This clock is determined by the ordered unparameterized trace. Section 3.3 justifies continuous Loewner driving of affine images; hence they admit this clock. The clock has infinite total lifetime almost surely: its total half-plane capacity T is positive, while spatial scale invariance makes its law equal to that of $r^2 T$ for every $r > 0$; the only such probability laws on $[0, \infty]$ are supported on $\{0, \infty\}$. Thus $T = \infty$. All functions below are functions of the ordered unparameterized trace.

Use the standard Borel space of these capacity-parameterized continuous curves, with the sigma-field generated by restrictions to finite time intervals. Equivalently, this can be regarded as a Borel realization of the usual ordered unparameterized path space. For $t > 0$, let

$$\mathcal{F}_t = \sigma\{\gamma(s) : 0 \le s \le t\}, \qquad \mathcal{F}_{0+} = \bigcap_{t>0} \mathcal{F}_t.$$

The latter is the initial trace-germ sigma-field. It is unaffected by changing any part of the curve after a positive capacity time. The canonical time is used only to give an explicit measurable coding of this germ.

For $r > 0$, let $S_r\gamma$ be the spatially dilated trace, again put in the canonical clock. Capacity scaling gives the exact identity

$$(S_r\gamma)(t) = r\gamma(t/r^2). \tag{SG1}$$

In particular S_r is invertible with inverse $S_{1/r}$.

12.2 Dilation ergodicity

Lemma 12.1 (Brownian scaling is mixing)**.** *For a one-sided Brownian motion $B = (B_t)_{t\geq 0}$ and any fixed $r > 1$, the measure-preserving transformation*

$$(\mathcal{S}_r B)_t = r B_{t/r^2}$$

is mixing and hence ergodic.

Proof. Set $Z_s = e^{-s/2}B_{e^s}$ for $s \in \mathbb{R}$. This is a stationary centered Gaussian process with

$$\mathbb{E}[Z_s Z_u] = e^{-|s-u|/2}.$$

It is the stationary Ornstein–Uhlenbeck process. Under $\mathcal{S}_r$, it becomes $Z_{s-2\log r}$. Covariances between finite collections of coordinates separated by $n(2\log r)$ tend to zero, so the corresponding jointly Gaussian vectors become independent in the limit. Approximation by finite-coordinate cylinder functions gives mixing on the full path sigma-field. The correspondence is invertible almost surely: $B_t = \sqrt{t}\, Z_{\log t}$ for $t > 0$, with $B_0 = 0$. Thus mixing transfers back to Brownian scaling. The two-sided log-time process here is obtained from a one-sided Brownian path at times $t > 0$; no independent Brownian past is being added. □

Lemma 12.2 (Affine SLE_6 is dilation mixing)**.** *Let A be any invertible linear map carrying $\mathbb{H}$ onto itself and fixing its positive boundary orientation. Define*

$$\mu_A = \mathrm{Law}(A^{-1}\eta), \qquad \eta \sim \mathrm{SLE}_6(\mathbb{H}, 0, \infty),$$

with the physical half-plane-capacity representative understood. For every fixed $r > 1$, S_r preserves μ_A and is mixing under μ_A.

Proof. The SLE trace is a measurable, scaling-equivariant function of its Brownian Loewner driver $\sqrt{6}B$. The trace exists almost surely. Therefore its dilation action is a factor of the Brownian scaling action and is mixing by Lemma 12.1. The map on ordered traces $\eta \mapsto A^{-1}\eta$ commutes with scalar spatial dilations, and is an invertible measurable map after recoding by capacity. Mixing and invariance therefore transfer to μ_A. □

12.3 Separating distinct metrics from arbitrarily short initial segments

Normalize the maps preserving $\mathbb{H}$ by

$$A_{b,c} = \begin{pmatrix} 1 & b \\ 0 & c \end{pmatrix}, \qquad b \in \mathbb{R},\ c > 0. \tag{SG2}$$

This parametrizes all constant positive-definite conformal metrics, up to a positive scalar, in a fixed half-plane frame: the tensor is $A_{b,c}^{\mathsf{T}} A_{b,c}$ with its $(1,1)$ entry normalized to 1. The laws $\mu_{A_{b,c}}$ are distinct. Indeed, the strictly increasing SLE_6 left-passage function F gives

$$\mu_{A_{b,c}}[L(x+iy)] = F\left(\frac{x+by}{cy}\right). \tag{SG3}$$

Equality of the probabilities at $z = i$ and $z = 1 + i$ first identifies b/c and then $1/c$, hence both parameters.

Lemma 12.3 (Unit-time restrictions distinguish the metrics)**.** *If $A \neq A'$ are two normalized maps in (SG2), the laws of their traces restricted to capacity time $[0,1]$ are distinct.*

Proof. If the restrictions agreed, dilation invariance and (SG1) would make the restrictions to $[0,T]$ agree for every $T > 0$. Those restrictions generate the full path sigma-field, so the entire laws would agree, contrary to (SG3). □

Theorem 12.4 (Metric decoding from an initial trace germ)**.** *Let $\{A_j : j \in J\}$ be any finite or countable collection of distinct normalized maps (SG2). There exists an $\mathcal{F}_{0+}$-measurable function D with values in $J \cup \{\dagger\}$ such that*

$$\mu_{A_j}[D(\gamma) = j] = 1 \qquad \text{for every } j \in J. \tag{SG4}$$

In particular, restrictions of these affine SLE_6 measures to the initial trace-germ sigma-field are mutually singular, even though each separately has its usual almost-sure germ zero-one property.

Proof. For each pair $j \neq k$, Lemma 12.3 supplies a Borel set of unit-time traces whose indicator f_{jk} has distinct expectations under μ_{A_j} and μ_{A_k}. There are only countably many such pairs; list all the resulting bounded unit-time observables as $(f_m)_{m\geq 1}$. Write $v_{j,m} = \int f_m \, d\mu_{A_j}$. The vectors $v_j = (v_{j,m})_m$ are pairwise distinct.

Fix $r > 1$. By Lemma 12.2 and Birkhoff's ergodic theorem, for each m and each j,

$$\frac{1}{N}\sum_{n=1}^{N} f_m(S_{r^n}\gamma) \longrightarrow v_{j,m} \qquad \mu_{A_j}\text{-almost surely.} \tag{SG5}$$

The intersection of these events over m still has full μ_{A_j} measure. Every term $f_m(S_{r^n}\gamma)$ depends only on the trace up to time r^{-2n}. For any $t > 0$, all sufficiently late terms are consequently $\mathcal{F}_t$-measurable. Omitting finitely many initial terms does not change a Cesàro limit. Thus every limit in (SG5), and the event that the limit exists, is $\mathcal{F}_{0+}$-measurable.

Define $D = j$ if all limits exist and their vector equals v_j; use $D = \dagger$ otherwise. Countability of J and pairwise distinctness of the v_j make this a measurable unambiguous definition. Formula (SG5) proves (SG4). □

12.4 A common decoder for the continuum of normalized metrics

The countable proof extends to all parameters $(b,c) \in \Theta = \mathbb{R} \times (0,\infty)$, provided the standard path coding is used jointly in the affine parameter. We state this coding point explicitly.

Joint Borel capacity coding. Let η be an ordinary capacity-parameterized SLE_6 trace and K_s its filled hull. Before recoding, its affine image is the continuous path $A_{b,c}^{-1}\eta(s)$. Its physical capacity clock is

$$u_{b,c,\eta}(s) = \tfrac{1}{2}\,\mathrm{hcap}(A_{b,c}^{-1}K_s). \tag{SG6}$$

Use the standard measurable coding of compact half-plane hulls, in which filling a continuous path prefix and taking its half-plane capacity are Borel operations. Linear transformation is jointly continuous on compact sets. Consequently $(b, c, \eta, s) \mapsto u_{b,c,\eta}(s)$ is Borel. The inverse clock is measurable because, on the set where the clock is continuous and strictly increasing,

$$u_{b,c,\eta}^{-1}(t) = \inf\{s : u_{b,c,\eta}(s) > t\}$$

can be computed using rational s (with equality interpreted by continuity). Thus evaluation of the recoded image on $[0, 1]$ is jointly Borel in (b, c, η). The good-clock conditions are Borel using rational tests; on their complement use a cemetery path. For each fixed (b, c) this complement has SLE probability zero by the affine local-growth and lifetime arguments above. No single full-probability set valid simultaneously for all affine parameters is required.

This paragraph uses the usual deterministic measurability of hull capacity as part of the standard Borel curve coding. If that coding fact is not taken as established, the next corollary should be read conditionally on joint Borel recoding. The preceding countable theorem needs only measurable recoding for each of its countably many fixed maps.

Corollary 12.5 (Continuum metric decoder)**.** *With the joint Borel coding just specified, there exists one $\mathcal{F}_{0+}$-measurable map*

$$D_{\mathrm{all}} : \mathcal{C} \longrightarrow \Theta \cup \{\dagger\}$$

such that for every $(b, c) \in \Theta$,

$$\mu_{A_{b,c}}[D_{\mathrm{all}}(\gamma) = (b, c)] = 1. \tag{SG7}$$

The same single measurable rule therefore recovers every normalized affine metric, with almost-sure correctness under each metric law separately.

Proof. Choose a countable generating π-system for the standard Borel space of unit-capacity path segments, including the cemetery state. For example, use finite intersections of rational-time evaluation cylinders in rational rectangles, together with the cemetery singleton. Let $(f_m)_{m\ge1}$ be their indicator functions. They determine probability laws on unit segments. Joint Borel recoding and integration against the fixed ordinary SLE law imply that

$$v : \Theta \to [0, 1]^{\mathbb{N}}, \qquad v_m(b, c) = \int f_m \, d\mu_{A_{b,c}}, \tag{SG8}$$

is Borel. Lemma 12.3 and the determining property of (f_m) imply that v is injective. Both parameter and sequence spaces are standard Borel, so the Lusin–Souslin theorem makes $v(\Theta)$ Borel and $v^{-1} : v(\Theta) \to \Theta$ Borel.

Fix $r > 1$. Form the vector $L(\gamma)$ of Cesàro limits

$$L_m(\gamma) = \lim_{N\to\infty} \frac{1}{N} \sum_{n=1}^{N} f_m(S_{r^n}\gamma),$$

using a cemetery value if any limit fails to exist. The proof of Theorem 12.4 shows that L is $\mathcal{F}_{0+}$-measurable. Define $D_{\mathrm{all}} = v^{-1}(L)$ when $L \in v(\Theta)$ and set it to $\dagger$ otherwise. This is a measurable definition; no measurable selection theorem is needed. For each fixed (b, c), ergodicity and Birkhoff applied to the countable family (f_m) give $L = v(b, c)$ almost surely under $\mu_{A_{b,c}}$. This proves (SG7). □

Quantifiers and limitations. The statement is one common decoder with correctness under each law: $\forall(b,c)\ \mu_{A_{b,c}}(D_{\text{all}} = (b,c)) = 1$. It does not assert one common full-probability set of Brownian seeds on which correctness holds simultaneously after every uncountable collection of affine transformations. More importantly, the decoded object is still an ordered initial trace germ at a smooth start, not the bare boundary germ of a random slit-domain tip.

What has and has not been proved. The theorem demonstrates an exact mechanism by which a microscopic ordered path germ can store a constant affine metric. It is more than identifiability of the law: a single arbitrarily short trace germ distinguishes any prescribed countable collection of metrics almost surely, and the continuum corollary extends this under the stated joint Borel coding. The result does not contradict a germ zero-one law, since that law concerns one probability measure, while the distinguishing full-measure germ sets differ between measures.

A hypothetical source-germ decoration could choose a metric from the smooth initial source geometry, then attempt to recover the same metric from the slit boundary at all later tips. That recovery does *not* follow from Theorem 12.4. Three obstacles remain:

1. A later tip is not a smooth half-plane starting boundary point; the nearby past trace can have a different two-sided or conditioned local law.

2. The remaining marked domain gives an embedded boundary germ, which may lose chronological information when the trace has touched itself. The theorem uses an oriented initial trace germ and does not prove that the needed data are recoverable from that boundary alone.

3. A domain Markov construction would need one measurable rule valid at all relevant stopping times and compatible with locality deletions, not merely a decoder at a deterministic initial point under each fixed metric law.

Extending local metric decoding to random interior or tip germs, or proving that such decoding is impossible from the domain state, is a concrete next research question. No full counterexample satisfying target consistency is claimed from this mechanism.

13 A fixed metric follows inside the affine-SLE ansatz with target consistency and reversal

Status. This theorem assumes that every marginal is already an affine image of SLE_6. It does not derive that ansatz from coarse percolation properties. It shows that quite weak physical coherence conditions then eliminate all variation of the affine conformal structure between domains and marked points. Geometric reversal is an explicit additional hypothesis; it is not asserted here to follow from RSW or from the original three axioms.

Let $\mathcal{D}$ be the class of bounded Jordan domains whose boundaries consist of finitely many regular real-analytic arcs meeting at corners. Each analytic arc extends to a neighborhood of its closed parameter interval. Restrict marks to regular boundary points, meaning interior points of these arcs. Suppose

$$P_{D,p,q} = (A_{D,p,q}^{-1})_* \,\mathrm{SLE}_6(A_{D,p,q}D, A_{D,p,q}p, A_{D,p,q}q), \tag{AA1}$$

where $A_{D,p,q}$ is an invertible orientation-preserving linear map. Write $[Q_{D,p,q}]$ for the projective positive-definite tensor $[A_{D,p,q}^{\mathsf{T}} A_{D,p,q}]$; brackets identify positive scalar multiples.

The following elementary local identifiability principle will be used: if two laws of the form (AA1), on domains agreeing near the same regular start, have identical initial paths stopped in

every sufficiently small ball, then their projective tensors are equal. Indeed blow up at the regular start. Conformal kernel convergence for ordinary SLE gives half-plane limits $A_i^{-1}\,\mathrm{SLE}_6(A_iH,0,\infty)$. Equality of local prefixes gives equality of these half-plane laws; the strictly monotone SLE_6 left-passage function makes $A_2A_1^{-1}$ a similarity. This argument uses only local SLE convergence and the affine ansatz.

Theorem 13.1 (Conditional global metric rigidity)**.** *Assume (AA1), together with the following properties on $\mathcal{D}$:*

1. Translation covariance.
2. Locality for nested domains*: if $D' \subset D$ agrees with D near regular marks p, q, the two initial laws agree until they encounter the removed set.*
3. Local target consistency*: for fixed D, p and any two regular targets q_1, q_2, the two initial laws agree when stopped on every sufficiently small circle centered at p.*
4. Geometric reversibility*: reversing an ordered path with law $P_{D,p,q}$ gives law $P_{D,q,p}$.*

Then there is one fixed positive-definite tensor Q_0, up to scalar, with $[Q_{D,p,q}] = [Q_0]$ for every admissible triple. Equivalently, one fixed linear map transforms the entire family into ordinary chordal SLE_6. No domain Markov hypothesis is needed for this conditional conclusion.

Proof. First, remove dependence on marks inside one domain. Fix D and a regular start p. Local target consistency and local identifiability make $[Q_{D,p,q}]$ independent of q. Ordinary SLE_6 is geometrically reversible by [3]. Therefore reversal of (AA1) retains its same tensor, and the assumed reversal of P implies

$$[Q_{D,p,q}] = [Q_{D,q,p}].$$

Given regular p, p', choose a third regular point q. Changing the target with fixed start q and reversing at the two ends shows

$$[Q_{D,p,q}] = [Q_{D,q,p}] = [Q_{D,q,p'}] = [Q_{D,p',q}].$$

Thus there is a tensor $[Q(D)]$ independent of both marks.

Second, compare nested domains with retained regular boundary arcs. If $D' \subset D$ agrees with D near two distinct regular boundary points, choose them as p, q. Locality makes the initial laws agree in some ball about p, and local identifiability gives $[Q(D')] = [Q(D)]$. The removed set may meet other parts of the boundary; it stays a positive distance from the selected marks.

Third, create a horizontal boundary segment without changing the metric. For any $D \in \mathcal{D}$, choose a horizontal line $\{y = c\}$ with $\min_{\partial D} \operatorname{Im} z < c < \max_{\partial D} \operatorname{Im} z$, at a regular value of the boundary height and not through a corner, and intersect D with $\{y > c\}$. Such a regular value with nonempty intersection exists: on each nonhorizontal analytic boundary arc the critical points are isolated, and horizontal arcs contribute only finitely many excluded height values. Choose the level also away from all corner heights. The analytic arcs are understood to extend regularly to neighborhoods of their closed parameter intervals, so there are only finitely many critical points unless an arc is horizontal. There are finitely many transverse crossings, and a component U of the intersection is again a piecewise real-analytic Jordan domain. It contains an original analytic boundary arc above the cutting line and also an open horizontal segment of the cutting line on its boundary. Choose two regular marks on the retained original arc. The second step gives $[Q(U)] = [Q(D)]$.

Choose a sufficiently short subsegment of the horizontal boundary segment. An upper half-disk V on that diameter lies in U, agrees with U near two distinct interior diameter points, and belongs to $\mathcal{D}$. The second step again gives $[Q(V)] = [Q(U)]$.

Finally, compare all horizontal upper half-disks. Translation covariance identifies their tensors for equal radii and arbitrary centers. Two upper half-disks with the same center and radii $r < R$ are nested and agree near any two distinct points on the smaller open diameter. The second step makes their tensors equal. Thus all such half-disks have one tensor $[Q_0]$, and the third step shows $[Q(D)] = [Q_0]$ for every D.

Choose a fixed A_0 with $A_0^{\mathsf{T}} A_0 = Q_0$. Equality of projective tensors gives $A_{D,p,q} = c_{D,p,q} R_{D,p,q} A_0$ with $c_{D,p,q} > 0$ and $R_{D,p,q}$ orthogonal. Similarity covariance of SLE_6 in (AA1) removes those extra factors, leaving exactly $(A_0^{-1})_* \,\mathrm{SLE}_6(A_0 D, A_0 p, A_0 q)$. □

Boundary regularity in the cutting step. A transverse horizontal cut of a piecewise real-analytic Jordan domain has finitely many intersections with its boundary. Each component is simply connected: a Jordan loop in the intersection has its interior in both the original Jordan domain and the upper half-plane. Its boundary consists of finitely many original arcs and cutting-line segments meeting at transverse corners, so the component is Jordan with piecewise real-analytic boundary. At least one retained original arc and one cutting segment occur in every nontrivial component chosen in the proof. This also makes all marked-point comparisons legitimate within the stated class.

Use for the percolation program. Local target consistency can be transferred from common-color discrete couplings to simultaneous marginal curve limits without random-domain kernel convergence, as established separately. Reversibility is a further assumption. For a general lattice model with critical density different from 1/2, reversing the color-oriented exploration is not automatically the same as swapping its start and target in a fixed convention: it also changes which boundary arc has each color. No exact discrete color symmetry, nor limiting reversal theorem, is supplied by the present argument. The main unsolved analytic step remains establishing (AA1), or an alternative local structure theorem, from the original axioms plus appropriate coherence.

14 Continuous affine labels: deterministic support and source independence

Status. The result here is conditional on the deterministic affine-SLE ansatz and an explicit continuity condition on its metric labels. It does not apply to a general exploration ensemble. It gives a further restriction inside that ansatz: a continuous conserved label cannot encode information in the source prime end. Source independence alone gives no target or domain independence; the final corollary adds explicit target-consistency and locality hypotheses to obtain a global metric.

14.1 The precise continuity premise

Let $T(D, p, q)$ be the determinant-one conformal metric tensor of the affine label, as in (RA6), on all bounded simply connected domains with two distinct prime ends. We use the following sufficient premise, stated in a form that is stable under the fixed affine change of coordinates needed for each starting law.

Affine-stable marked-chart continuity. For every fixed invertible real-linear map L, suppose $f_n : \mathbb{H} \to E_n$ are conformal maps onto bounded simply connected domains, with a nondegenerate

conformal limit $f_n \to f : \mathbb{H} \to E$ locally uniformly in $\mathbb{H}$. The marked prime ends are $f_n(0), f_n(\infty)$ and $f(0), f(\infty)$, respectively; these are prime-end extensions, whether or not they have Euclidean point limits. Require

$$T(LE_n, Lf_n(0), Lf_n(\infty)) \longrightarrow T(LE, Lf(0), Lf(\infty)). \tag{SS1}$$

Thus the topology is locally uniform convergence of inverse conformal charts with the two designated boundary preimages $0, \infty$, and the premise also applies after each fixed affine map. This is an explicit additional hypothesis; it is not inferred merely from DMP, locality, or physical translation invariance. An equivalent affine-stable marked kernel notion may be used instead, provided it supplies (SS1).

Why fixed affine maps preserve this marked topology (proved using standard Beltrami stability). In fact the displayed premise for all L follows from its case $L = I$. Here is the needed topological fact. Write an orientation-preserving real-linear map as $L(z) = \alpha z + \beta \bar{z}$, with $|\beta/\alpha| < 1$. The Beltrami coefficients of $L \circ f_n$ are

$$\mu_n(z) = \frac{\beta}{\alpha} \frac{\overline{f_n'(z)}}{f_n'(z)}.$$

Since the nonvanishing derivatives f_n' converge locally uniformly to f', these coefficients converge pointwise in $\mathbb{H}$ to the analogous μ, all with the same bound $k < 1$. Extend them to the lower half-plane by reflection, $\mu_n(\bar{z}) = \overline{\mu_n(z)}$. Let h_n be the normalized quasiconformal sphere solutions fixing $0, 1, \infty$. The standard stability theorem for the measurable Riemann mapping theorem (uniform bound $k < 1$ and almost-everywhere convergence of coefficients) gives locally uniform convergence of h_n and their inverses to h and h^{-1}. Reflection symmetry and normalized uniqueness imply that every h_n maps $\mathbb{H}$ onto itself and fixes the two marked boundary points $0, \infty$.

The maps

$$\chi_n = L \circ f_n \circ h_n^{-1} : \mathbb{H} \longrightarrow LE_n$$

are conformal by Beltrami factorization. They converge locally uniformly to the nondegenerate conformal map $\chi = L \circ f \circ h^{-1}$, and their marked endpoints are precisely $Lf_n(0), Lf_n(\infty)$. This proves continuity of a fixed orientation- preserving affine change of the physical domains in the marked conformal- chart topology. A reflection handles the other orientation. Consequently ordinary marked-chart continuity of T implies (SS1). Keeping (SS1) explicit in the proof below makes clear that its charts are holomorphic in the affine SLE coordinates, even when their physical pullbacks are only quasiconformal. The additional analytical input in this paragraph is the standard normalized Beltrami stability theorem; no claim about Euclidean uniform boundary convergence is needed.

14.2 Full support transfers conservation to deterministic Loewner controls

Fix a starting label $A = A(D, p, q)$ and a conformal map $f : \mathbb{H} \to AD$ with $f(0) = Ap$, $f(\infty) = Aq$. For any continuous real function w on $[0, t]$ with $w(0) = 0$, let g_t^w denote the associated hydrodynamic Loewner map on its surviving half-plane domain. Define the marked target state by the inverse conformal chart

$$f_t^w(z) = f\big((g_t^w)^{-1}(z + w(t))\big), \qquad z \in \mathbb{H}. \tag{SS2}$$

This state is a simply connected domain with two designated prime ends for *every* continuous w. The assertion does not say that every such w generates a trace. No trace is needed to define the

domain state in (SS2). For the Brownian driving paths of SLE the state is its actual remaining target domain.

The map $w \mapsto f_t^w$ is continuous from the uniform norm on $C[0,t]$ to locally uniform convergence of conformal maps on $\mathbb{H}$. For example, the inverse map can be represented by the reverse ODE

$$\dot{v}_s = -\frac{2}{v_s - w(t-s)}, \quad v_0 = z, \quad (g_t^w)^{-1}(z) = v_t. \tag{SS3}$$

Its imaginary part is nondecreasing. On a compact set with $\operatorname{Im} z \geq \delta > 0$, denominators stay away from zero and the usual ODE stability estimate gives locally uniform continuity in w. Translations by $w(t)$ and composition with f preserve this continuity. The corresponding metric functional

$$\mathcal{T}_t(w) = T\big(A^{-1} f_t^w(\mathbb{H}), A^{-1} f_t^w(0), A^{-1} f_t^w(\infty)\big) \tag{SS4}$$

is therefore continuous by (SS1), with $L = A^{-1}$.

Deterministic-support lemma (proved). If the affine-label ensemble has strong DMP and satisfies (SS1), then for every $t > 0$ and every continuous control $w \in C[0,t]$ with $w(0) = 0$,

$$\mathcal{T}_t(w) = T(D, p, q). \tag{SS5}$$

For the source-independence conclusion below it suffices to apply this to smooth linear ramps.

Proof. Under the original affine SLE law the driving process in AD is $\sqrt{6}B$. The DMP conservation criterion (RA7) gives $\mathcal{T}_t(\sqrt{6}B) = T(D, p, q)$ almost surely. Brownian measure has full support on the space of continuous functions starting at zero in the uniform topology. If (SS5) failed at one control w, continuity of $\mathcal{T}_t$ would give an open uniform neighborhood on which it differs from the fixed tensor. Brownian full support gives that neighborhood positive probability, contradicting almost-sure conservation. □

One could avoid defining the functional at non-trace-generating controls by restricting the statement to controls at which the state functional has a continuous extension, and applying full support there. In particular smooth ramps suffice. The all-prime-end ansatz already makes (SS4) defined on every state, so the formulation above is simpler.

14.3 Rapid driving ramps move the source at vanishing capacity

Theorem (proved, within the stated ansatz). Under strong DMP and (SS1), the tensor $T(D, p, q)$ is independent of the source prime end: for every bounded simply connected D and every three prime ends with $p, p' \neq q$,

$$T(D, p, q) = T(D, p', q). \tag{SS6}$$

It may still depend on the target or on the domain; these are unresolved by this argument.

Proof. Use the notation above and write $Ap' = f(a)$ for the unique finite $a \in \mathbb{R}$ corresponding to the new source. Choose $t_n \downarrow 0$ and use the smooth control

$$w_n(s) = \frac{as}{t_n}, \qquad 0 \leq s \leq t_n.$$

The deterministic-support lemma says that every resulting metric equals $T(D, p, q)$. From the reverse equation (SS3),

$$|(g_{t_n}^{w_n})^{-1}(z) - z| \leq \frac{2t_n}{\operatorname{Im} z}. \tag{SS7}$$

Consequently, locally uniformly in $\mathbb{H}$,

$$f_{t_n}^{w_n}(z) = f\big((g_{t_n}^{w_n})^{-1}(z+a)\big) \longrightarrow f(z+a).$$

The limiting chart has image AD, source $f(a) = Ap'$, and the same target $f(\infty) = Aq$. Applying (SS1) with $L = A^{-1}$ gives (SS6). □

The controls may have very large slopes as $t_n \to 0$; this is harmless. Each fixed ramp lies in the support of Brownian driving on its own finite interval. The argument uses the possibility of rare events, not their uniform probability as $n \to \infty$. The limiting source change is a statement about marked prime-end charts, not Euclidean convergence of a curve of vanishing diameter: the capacity vanishes while the tip may travel far along a thin boundary region.

14.4 A fixed metric in the affine ansatz without a reversal hypothesis

Corollary (proved, conditional on the global affine ansatz). Suppose the ensemble has a deterministic affine-SLE label on *all* bounded simply connected domains and distinct prime ends, has strong DMP, and its normalized tensor T is continuous for marked inverse-conformal- chart convergence. Suppose in addition that its piecewise real-analytic Jordan subfamily has translation covariance, locality for nested domains retaining neighborhoods of regular marks, and local target consistency in the sense used by the conditional global metric-rigidity theorem. Then there is one fixed determinant-one tensor Q_0 such that

$$T(D, p, q) = Q_0$$

for every bounded simply connected marked domain. No geometric reversal hypothesis is needed for this conditional conclusion.

Proof. The source-independence theorem removes dependence on p. On a regular piecewise real-analytic Jordan domain, local target consistency together with the local half-plane identifiability argument removes dependence on q as well. Thus these domains have a tensor $Q(D)$ independent of both marks. The nested-cut argument in the preceding conditional global metric-rigidity theorem now applies verbatim, without its reversal step: nested locality identifies tensors of domains agreeing near two retained regular marks; a transverse horizontal cut creates a horizontal segment; a small upper half-disk on that segment retains the metric; translation and nesting identify all such half-disks. Hence all piecewise real-analytic Jordan domains have the same tensor Q_0.

To extend to an arbitrary bounded simply connected D, choose a conformal $f : \mathbb{D} \to D$ and distinct circle points ζ_p, ζ_q representing p, q. For $r_n \uparrow 1$, set

$$D_n = f(r_n\mathbb{D}), \qquad p_n = f(r_n\zeta_p), \quad q_n = f(r_n\zeta_q).$$

Each D_n is an analytic Jordan domain: f extends holomorphically and univalently across the compact circle $|z| = r_n$, with nonzero derivative. Choose a conformal $m : \mathbb{H} \to \mathbb{D}$ with $m(0) = \zeta_p$, $m(\infty) = \zeta_q$. The marked inverse charts

$$f_n(z) = f(r_n m(z))$$

converge locally uniformly to $f(m(z))$, giving exactly the desired marked prime-end limit, even if f has no Euclidean limit at one of the selected circle points. Tensor continuity and $T(D_n, p_n, q_n) = Q_0$ give $T(D, p, q) = Q_0$. The fixed-affine normalizer follows from SLE similarity covariance as in the preceding theorem. □

This corollary removes a reversal assumption *inside the global affine ansatz*, at the price of strong DMP and a precise continuity premise. It still supplies no argument that the original three axioms, RSW, or local target consistency force a marginal exploration law to belong to that ansatz.

15 Unsuccessful routes, revised objectives, and next experiments

Why several tempting arguments fail. Independent site colors give exact discrete domain Markov and locality at every occupation probability; these axioms do not by themselves encode criticality. Near-critical scaling limits are therefore natural candidates, but configuration convergence and a Markov theorem in the near-critical parameter do not establish the required ordered exploration kernels at random slit domains. We do not use this incomplete transfer as the main counterexample. Nor does singularity relative to isotropic SLE alone exclude an anisotropic affine image. A genuine dilation defect would do so.

A random mixture over fixed affine metrics is also not automatically domain Markov: observing the initial curve can update the metric posterior. The exact example avoids that failure through deterministic terminal information that remains visible in every current domain. This should be distinguished from unsupported claims that every stationary random environment yields a domain-Markov annealed interface.

Applying Schramm's usual Brownian-driving argument directly would silently assume that a conformally mapped slit domain has the same law as the original domain. Physical translations do not imply that premise. The deletion-chart identity above identifies the precise missing drift control. We have not proved that the original axioms imply semimartingale driving, driving martingales, or even a fixed underlying conformal structure.

A revised research question, not a proved classification. Restrict to nondegenerate continuous Loewner exploration laws with strong domain Markov, fixed-target locality, physical translation covariance, consistency under changing distant boundary conditions and target, and continuity under controlled Jordan-domain approximation, including changes near the terminal point. Require a genuine criticality condition, such as nondegenerate crossing estimates uniform over all scales, formulated for a common continuum coloring or a compatible exploration system. Does this force one deterministic conformal metric, and then affine SLE_6? Even the sufficiency of this strengthened collection is open here. One must not promote its plausibility to a theorem. Full similarity covariance alone is already defeated by the exact construction above.

Concrete priorities for the next session. The local common-coloring transfer is now proved in Section 11. The next limit questions are controlled marked-domain stability and conditional-kernel convergence in random slit domains, ideally from uniform crossing estimates. The main structural question is to derive affine-SLE marginals, or a different useful local structure theorem, for arbitrary exploration laws. Once that ansatz is available, the conditional common-metric theorem above removes the domain dependence under explicit coherence and continuity hypotheses. A parallel route is to seek an observable implying the driving-local- martingale premise in Section 7, possibly after one affine normalization. That premise would finish identification by the proved conditional theorem, but deriving it from coarse percolation properties remains open. Finally examine whether later accessible frontier germs can encode geometry despite target consistency: initial ordered-germ decoding is now established, but the later unlabelled-boundary construction is not. No full additional counterexample is claimed.

Reproducibility and preservation. The concise project handoff is `STATUS.md`. The polished counterexample and global-deformation results are presented separately in `final.tex`. Independent scratch investigations and compilation products are under `work/`; this record contains the actual mathematical arguments and does not require the conversation or those scratch files to be understood. Successful LaTeX compilation checks presentation only. It is not evidence that a mathematical proof is correct.

References

[1] G. F. Lawler, *Schramm–Loewner Evolution*, Park City lecture notes (2007), especially Lectures 3–4. https://math.uchicago.edu/~lawler/utah.pdf.

[2] Y. Yuan, *Topological characterisations of Loewner traces*, Indiana Univ. Math. J. **71** (2022), 1027–1046, Theorem 1.1 and Remark 1.4. https://arxiv.org/abs/2003.05535.

[3] J. Miller and S. Sheffield, *Imaginary geometry III: reversibility of SLE_κ for $\kappa \in (4, 8)$*, Ann. of Math. **184** (2016), 455–486. https://annals.math.princeton.edu/2016/184-2/p03.

[4] P. Nolin and W. Werner, *Asymmetry of near-critical percolation interfaces*, J. Amer. Math. Soc. **22** (2009), 797–819. https://arxiv.org/abs/0710.1470.

[5] C. Garban, G. Pete, and O. Schramm, *The scaling limits of near-critical and dynamical percolation*, J. Eur. Math. Soc. **20** (2018), 1195–1268, especially Section 13.2. https://arxiv.org/abs/1305.5526.